

PMath 348: Fields and Galois Theory

We will spend weeks 1-3 on more ring theory, weeks 4-6 on field theory, week 7-10 on Galois theory, weeks 11-12 on applications.

Chapter 1: Introduction and Review of Ring Theory

1.1 Introduction to Galois Theory

In middle school, we saw linear equations of the form $ax + b = 0$, with $a, b \in \mathbb{R}$ and $a \neq 0$. Its unique solution is $x = \frac{-b}{a}$. Then, in high school, we saw quadratic equations of the form $ax^2 + bx + c = 0$ with $a, b, c \in \mathbb{R}$ and $a \neq 0$. Then we know its only two solutions are $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. We see this solution contains only addition, subtraction, multiplication, division, and taking square roots.

Definition

An expression involving only addition, subtraction, multiplication, division, and taking n th roots is called a radical.

There's also the cubic formula (after Tartagila, del Ferro, and Fontana). They showed all cubic equations can be reduced to $x^3 + px = q$. Then, a solution of the above equation is given by

$$x = \sqrt[3]{\frac{q}{2} + \sqrt{\frac{p^3}{27} + \frac{q^2}{4}}} + \sqrt[3]{\frac{q}{2} - \sqrt{\frac{p^3}{27} + \frac{q^2}{4}}}$$

There's also a solution to the quartic (see Bonus Question 1).

But what about the quintic? This question was attempted by Euler, Bezout, and Lagrange without success. In 1799, Ruffini gave a 516 page proof about the unsolvability of the quintic equation by radicals. His proof was "almost right", and in 1824, Abel filled in the gap in Ruffini's proof.

So quintics aren't generally solvable by radicals, but some are (we can construct obvious ones). Can we answer this question for each equation?

Question

Given a quintic equation, is it solvable by radicals?

The trick of Galois theory is to ask the reverse question.

Reverse Question

Suppose that a radical solution exists. What does its associated quintic equation look like?

This is how we'll do it:

Two main steps of Galois Theory

1. Link a root of a quintic equation, say α , to the smallest field containing $\mathbb{Q}$ and α , denoted $\mathbb{Q}(\alpha)$. But $\mathbb{Q}(\alpha)$ is a field and our knowledge about fields is limited. As an example, consider $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, the smallest field containing $\mathbb{Q}, \sqrt{2}, \sqrt{3}$. But between this field and $\mathbb{Q}$, there are a lot of fields lying strictly between them (like $\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3}), \mathbb{Q}(\sqrt{6}), \mathbb{Q}(\sqrt{2} + \sqrt{3}), \mathbb{Q}(\sqrt{2} - \sqrt{3})$).

2. Link the field $\mathbb{Q}(\alpha)$ to a group. More precisely, we associate the field extension $\mathbb{Q}(\alpha)$ to $\mathbb{Q}$ to the group $\text{Aut}_{\mathbb{Q}}(\mathbb{Q}(\alpha)) = \{\psi : \mathbb{Q}(\alpha) \rightarrow \mathbb{Q}(\alpha) \text{ is an isomorphism and } \psi|_{\mathbb{Q}} = 1\}$. One can show if α is "good", say "algebraic", then $\text{Aut}_{\mathbb{Q}}(\mathbb{Q}(\alpha))$ is finite. Furthermore, there is a one-to-one correspondence between the intermediate fields between $\mathbb{Q}(\alpha)$ and $\mathbb{Q}$ and the subgroups of $\text{Aut}_{\mathbb{Q}}(\mathbb{Q}(\alpha))$. The idea of Galois theory is to build this bridge.

1.2 Review of Ring Theory

Definition

A set R is a (unitary) ring if it has two operations, addition $+$ and multiplication $\cdot$, such that for all $a, b, c \in R$:

- (1) $a + b \in R$
- (2) $a + b = b + a$
- (3) $a + (b + c) = (a + b) + c$
- (4) There exists $0 \in R$ such that $a + 0 = a = 0 + a$
- (5) There exists $-a \in R$ such that $a + (-a) = 0 = (-a) + a$
- (6) $ab = a \cdot b \in R$
- (7) $a(bc) = (ab)c$
- (8) There exists $1 \in R$ such that $a1 = a = 1a$
- (9) $a(b + c) = ab + ac$ and $(b + c)a = ba + ca$

The ring is commutative if it satisfies:

- (10) $ab = ba$

In this class, all rings are commutative (and unitary).

Definition

Let R be a commutative ring. We say $u \in R$ is a unit if u has a multiplicative inverse, denoted by u^{-1} , i.e. $uu^{-1} = 1$. Let R^* denote the set of all units in R . Note that $(R^*, \cdot)$ is a group.

Definition

A commutative ring $R \neq \{0\}$ with $R^* = R \setminus \{0\}$ is a field.

Definition

A commutative ring $R \neq \{0\}$ is an integral domain if for $a, b \in R$ with $ab = 0$, then $a = 0$ or $b = 0$.

Examples

$\mathbb{Z}$ is an integral domain, $\mathbb{Q}, \mathbb{R}, \mathbb{C}$, and $\mathbb{Z}_p$ where p is prime are all fields.

Proposition 1.1

Every subring of a field (including the field itself) is an integral domain. ■

Definition

A subset I of a commutative ring R is an ideal if for $a, b \in I$ and $r \in R$, we have $a - b \in I$ and $ra \in I$.

Example

Let I be an ideal of a commutative ring R . If $1_R \in I$, then $I = R$.

Example

The only ideals of a field F are $\{0\}$ and F .

Properties of the ring of integers $\mathbb{Z}$

- $\mathbb{Z}$ is an integral domain, with units ± 1 .

-We also have a division algorithm in $\mathbb{Z}$: for $a, b \in \mathbb{Z}$ with $a \neq 0$, we can write $b = qa + r$, with $q, r \in \mathbb{Z}$ and $0 \leq r < |a|$.

-We can use the division algorithm to show that all ideals of $\mathbb{Z}$ are $I = \langle n \rangle = n\mathbb{Z}$ for some $n \in \mathbb{Z}$, and thus the generator is unique if we insist $n > 0$.

-Consider all fields containing $\mathbb{Z}$. Their intersection (the smallest field containing $\mathbb{Z}$) is $\mathbb{Q} = \{\frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0\}$.

Properties of the ring $F[x]$ (F a field)

Let F be a field. Define $F[x] = \{f(x) = a_0 + a_1x + \cdots + a_mx_m : a_i \in F, m \in \mathbb{N}\}$.

- If $a_m = 1$, the polynomial is monic.
- If $a_m \neq 0$, we define the degree of f to be $\deg f = m$. We further define $\deg(0) = -\infty$.
- For $f(x), g(x) \in F[x]$, we have $\deg(f(x)g(x)) = \deg f(x) + \deg g(x)$.
- $F[x]$ is an integral domain.
- The units of $F[x]$ are $F^* = F \setminus \{0\}$.
- $F[x]$ has a division algorithm: for $f(x), g(x) \in F[x]$ with $f(x) \neq 0$, we can write $g(x) = q(x)f(x) + r(x)$, $q(x), r(x) \in F[x]$ with $\deg(r(x)) < \deg(f(x))$.
- We can use the division algorithm to prove all ideals I of $F[x]$ are of the form $I = \langle f(x) \rangle$, where we can make it unique by insisting that $f(x)$ be monic.
- Consider all fields containing $F[x]$. Their intersection is the set of all rational functions $F(x) = \left\{ \frac{f(x)}{g(x)} : f(x), g(x) \in F[x], g(x) \neq 0 \right\}$.

Let I be an ideal of a ring R . We recall that the additive quotient group R/I (the quotient makes sense since R is an abelian group under addition, and thus its additive subgroup in I is normal) is a ring with the multiplication $(r + I)(s + I) = rs + I$. The 1 of R/I is $1 + I$.

Theorem 1.2 (First Isomorphism Theorem)

Let $\phi : R \rightarrow S$ be a ring homomorphism. Then the $\ker \phi$ is an ideal of R , and we have $R/\ker \phi \cong \text{im } \phi$. ■

Example

Let F be a field and S be a ring, and let $\phi : F \rightarrow S$ be a ring homomorphism. The kernel of ϕ is an ideal of F , so we only have 2 choices: $\ker \phi = \{0\}$ or F . Therefore, the only possible choices we have are that ϕ is the zero map or it is injective.

Definition

Let R be a commutative ring. An ideal $P \neq R$ of R is a prime ideal if whenever $r, s \in R$ such that $rs \in P$, then either $r \in P$ or $s \in P$.

Definition

Let R be a commutative ring. An ideal $M \neq R$ of R is a maximal ideal if whenever there is an ideal A of R such that $M \subseteq A \subseteq R$, we have $M = A$ or $A = R$.

Proposition 1.3

Every maximal ideal is prime. ■

Theorem 1.4

Let I be an ideal of a commutative ring R and $I \neq R$. Then

- (1) I is a maximal ideal if and only if R/I is a field.
- (2) I is a prime ideal if and only if R/I is an integral domain. ■

Chapter 2: Integral domains

2.1 Irreducibles and Primes

Definition

Let R be an integral domain and $a, b \in R$. We say a divides b , denoted by $a \mid b$, if $b = ca$ for some $c \in R$.

We recall that in $\mathbb{Z}$, if $n \mid m$ and $m \mid n$, then $n = \pm m$ and $\langle n \rangle = \langle m \rangle$. Also, in $F[x]$, if $f(x) \mid g(x)$ and $g(x) \mid f(x)$, then $f = cg$ for some $c \in F^*$ and $\langle f \rangle = \langle g \rangle$.

Proposition 2.1

Let R be an integral domain. For $a, b \in R$, the following are equivalent:

- (1) $a \mid b$ and $b \mid a$
- (2) $a = ub$ for some unit $u \in R$
- (3) $\langle a \rangle = \langle b \rangle$.

Proof:

Proving (1) implies (2), suppose $a \mid b$ and $b \mid a$, write $b = va$ and $a = ub$ for some $u, v \in R$. If $a = 0$, then $b = 0$ and thus $a = 1b$. Otherwise, if $a \neq 0$, then $a = u(va) = (uv)a$. Since this product is nonzero, by integrality, this implies that $uv = 1$. Therefore, u is a unit.

Proving (2) implies (3), if $a = ub$, then we have $\langle a \rangle \subseteq \langle b \rangle$. Since u is a unit and $b = u^{-1}a$, we have $\langle b \rangle \subseteq \langle a \rangle$. It follows that $\langle a \rangle = \langle b \rangle$.

Proving (3) implies (1), if $\langle a \rangle = \langle b \rangle$, then $a \in \langle b \rangle$, then $a = ub$ for some $u \in R$, so $b \mid a$, and similarly $b \in \langle a \rangle$ so $b = va$ for some $v \in R$, so $a \mid b$. ■

Definition

Given an integral domain R , and $a, b \in R$, we say a is associated to b , denoted by $a \sim b$, if $a \mid b$ and $b \mid a$. By Proposition 2.1, $\sim$ is an equivalence relation in R . One can show that

- (1) If $a \sim a'$, and $b \sim b'$, then $ab \sim a'b'$.
- (2) If $a \sim a'$, and $b \sim b'$, then $a \mid b$ if and only if $a' \mid b'$.

Example

Let $R = \mathbb{Z}[\sqrt{3}]$, which is an integral domain. Note that $(2 + \sqrt{3})(2 - \sqrt{3}) = 1$, thus $2 + \sqrt{3}$ is a unit in R . Since $3 + 2\sqrt{3} = (2 + \sqrt{3})\sqrt{3}$, we have $3 + 2\sqrt{3} \sim \sqrt{3}$.

Definition

Let R be an integral domain. We say $p \in R$ is irreducible if $p \neq 0$, not a unit, and if $p = ab$ with $a, b \in R$, then either a or b is a unit in R .

Example

Let $R = \mathbb{Z}[\sqrt{-5}]$ and $p = 1 + \sqrt{-5}$. We claim p is irreducible in R . For $d = m + n\sqrt{-5}$, the norm of f is defined to be $N(d) = m^2 + 5n^2$. One can show that $N(ab) = N(a)N(b)$ and $N(d) = 1$ if and only if d is a unit. Moreover, we see $N(1 + \sqrt{-5}) = 6$. Now suppose that $p = ab \in R$. Then $6 = N(p) = N(a)N(b)$, and note that $6 = (6)(1) = (2)(3)$. If $N(d) = m^2 + 5n^2 = 2$, with $m, n \in \mathbb{Z}$, looking mod 5, then $n = 0$. However, $m^2 \neq 2$, hence $N(d) \neq 2$, and an identical argument works for $N(d) \neq 3$. Hence either $N(a) = 1$ or $N(b) = 1$, i.e. either a or b is a unit in R . Thus p is irreducible.

Proposition 2.2

Let R be an integral domain with $p \neq 0 \in R$ and not a unit. The following are equivalent:

- (1) p is irreducible.
- (2) if $d \mid p$, then $d \sim 1$ or $d \sim p$.
- (3) If $p \sim ab$ in R , then $p \sim a$ or $p \sim b$.
- (4) If $p = ab$, then $p \sim a$ or $p \sim b$.

Proof:

Proving (1) implies (2), if $d \mid p$, then $p = ad$ for $a \in R$, so by (1), d or a is a unit, then respectively we have $d \sim 1$ or $d \sim p$ (the latter by our initial equation).

Proving (2) implies (3), suppose $p \sim ab$. Therefore, $b \mid p$. Thus, by (2), either $b \sim 1$ or $b \sim p$. In the former case, this implies $a \sim p$ since it differs from p by the product of two units (the original one from the $\sim$ and b).

Proving (3) implies (4), take the unit in the definition of $\sim$ to be 1.

Proving (4) implies (1), write $p = ab$, which by (4), gives $p \sim a$ or $p \sim b$. Without loss of generality suppose $p \sim a$, write $a = up$ for a unit u . Since R is commutative, $p = ab = (up)b = p(ub)$, so cancelling the p 's (we are in an integral domain), we get $ub = 1$. Therefore b is a unit. ■

Definition

Let R be an integral domain and $p \in R$. We say p is prime if $p \neq 0$, not a unit, and if $p \mid ab$ with $a, b \in R$, then $p \mid a$ or $p \mid b$. One can show that if $p \sim q$, p is prime if and only if q is prime.

Proposition 2.3

Let R be an integral domain and $p \in R$. If p is prime, then p is irreducible.

Proof:

Let $p \in R$ be prime. Suppose $p = ab$, therefore by definition we have $p \mid ab$ so $p \mid a$ or $p \mid b$. Without loss of generality, suppose $p \mid a$. Write $a = dp$ for $d \in R$. Since R is commutative, we have $a = dp = d(ab) = a(db)$, so cancelling the a 's, we have $db = 1$, so b is a unit. ■

Example

The converse of the above proposition is false. Consider $R = \mathbb{Z}[\sqrt{-5}]$, and let $p = 1 + \sqrt{-5} \in R$. We showed before that p is irreducible. We claim p is not prime. Indeed, note that $p(1 - \sqrt{-5}) = 6 = (2)(3)$. But if $p \mid 2$, then looking at the norms we must have that 6 divides 4, impossible, or if $p \mid 3$, then we must get 6 dividing 9, impossible.

Question

In $\mathbb{Z}$ the notions of an element being prime and irreducible are interchangeable (the set of primes). Similarly, in $\mathbb{F}[x]$ (the set of irreducible polynomials). So what are the additional properties in $\mathbb{Z}$ or $\mathbb{F}[x]$ that allows us to get the condition that all irreducibles are primes?

2.2 Ascending Chain Conditions

Definition

An integral domain R is said to satisfy the ascending chain condition on principal ideals (ACCP) if for any ascending chain $\langle a_1 \rangle \subseteq \langle a_2 \rangle \subseteq \dots$ of principal ideals in R , then there exists $n \in \mathbb{N}$ such that $\langle a_n \rangle = \langle a_{n+1} \rangle = \dots$.

Example

$\mathbb{Z}$ satisfies this condition. Indeed, suppose $\langle a_1 \rangle \subseteq \langle a_2 \rangle \subseteq \langle a_3 \rangle \subseteq \dots$. Then $a_2 \mid a_1$, $a_3 \mid a_2, \dots$. Taking absolute values gives $|a_1| \geq |a_2| \geq |a_3| \geq \dots$. Since each $|a_i| \geq 0$ is an integer, only finitely many of these inequalities can be strict, implying we can get $|a_n| = |a_{n+1}| = \dots$ for some $n \in \mathbb{N}$, implying $\langle a_n \rangle = \langle a_{n+1} \rangle = \dots$.

Example

Consider $R = \{n + xf : n \in \mathbb{Z}, f \in \mathbb{Q}[x]\}$, the set of rational polynomials with integer constant terms. This is an integral domain, but does not satisfy the ascending chain condition on principal ideals. Indeed, for the ideal $\langle x \rangle$, the degree 1 terms must have integer coefficients, but for the ideal $\langle \frac{1}{2}x \rangle$, the degree 1 terms have denominator 2, so $\langle x \rangle \subsetneq \langle \frac{1}{2}x \rangle$, and a similar argument shows $\langle x \rangle \subsetneq \langle \frac{1}{2}x \rangle \subsetneq \langle \frac{1}{4}x \rangle \subsetneq \dots \subsetneq \langle \frac{1}{2^n}x \rangle \subsetneq \dots$, so this ring does not have the desired property.

Theorem 2.4

Let R be an integral domain satisfying the ascending chain condition on principal ideals. If $a \in R$ with $a \neq 0$, not a unit, then a is a product of irreducible elements of R .

Proof:

Suppose $a \in R$, not a unit and nonzero, is not a product of irreducible elements, so it is not irreducible itself. Therefore, by Proposition 2.2, we can write $a = x_1 a_1$ where a is not associated to either x_1 or a_1 . Note that at least one of x_1 and a_1 , say a_1 , cannot be a product of irreducibles by a 's factorization (otherwise a would be a product of irreducibles). This gives $\langle a \rangle \subsetneq \langle a_1 \rangle$ since $a \in \langle a_1 \rangle$ but a is not associated to a_1 , and the same argument gives $a_2, a_3, \dots \in R$ such that $\langle a \rangle \subsetneq \langle a_1 \rangle \subsetneq \langle a_2 \rangle \subsetneq \dots$.

This contradicts the ascending chain condition. Therefore, a must be a product of irreducible elements. ■

Theorem 2.5

If R is an integral domain satisfying the ascending chain condition on principal ideals, then so too is $R[x]$.

Proof:

Suppose that $R[x]$ does not satisfy the ascending chain condition on principal ideals. Then there exists a chain of principal ideals $\langle f_1 \rangle \subsetneq \langle f_2 \rangle \subsetneq \dots$ in $R[x]$. Thus for all $i \in \mathbb{N}$, we have $f_{i+1} \mid f_i$. Let a_i denote the leading coefficient of each f_i . Fix an i , giving that there exists $g \in R[x]$ such that $f_i = f_{i+1}g$. Therefore, by looking at the leading coefficients, we have $a_i = a_{i+1}b$, where b is the leading coefficient of g . Therefore, $a_{i+1} \mid a_i$, giving us a chain of principal ideals in R : $\langle a_1 \rangle \subseteq \langle a_2 \rangle \subseteq \dots$. By the ascending chain condition, there exists $n \in \mathbb{N}$ such that $\langle a_n \rangle = \langle a_{n+1} \rangle = \dots$, so $a_n \sim a_{n+1} \sim \dots$. Now, for some $m \geq n$, let $f_m = gf_{m+1}$ for $g \in R[x]$, with b its leading coefficient. Then, we have $a_m = ba_{m+1}$. Because a_m and a_{m+1} are associated, b is a unit in R . However, $g(x)$ is not a unit and thus not a constant in $R[x]$ (otherwise f_m and f_{m+1} would generate the same ideals, violating $\langle f_m \rangle \subsetneq \langle f_{m+1} \rangle$). Therefore, $\deg f_m > \deg f_{m+1}$, which can only hold finitely many times because degrees are non-negative integers, so we cannot have an infinite decreasing sequence, which is a contradiction, because $\langle f_n \rangle \subsetneq \langle f_{n+1} \rangle \subsetneq \dots$ imply $\deg f_n > \deg f_{n+1} > \dots$. Therefore, the ascending chain condition on principal ideals must hold for $R[x]$. ■

Example

Since $\mathbb{Z}$ satisfies the ascending chain condition on principal ideals, so does $\mathbb{Z}[x]$, and by induction, any finite number of polynomial variables works too $\mathbb{Z}[x_1, \dots, x_n]$.

2.3 Unique Factorization Domains and Principal Ideal Domains

Definition

An integral domain R is called a unique factorization domain (abbreviated UFD) if it satisfies the following conditions:

- (1) If $a \in R$, $a \neq 0$ and not a unit, then a is a product of irreducible elements in R .
- (2) If $p_1 p_2 \dots p_r \sim q_1 q_2 \dots q_r$ where p_i and q_j are irreducible, then $r = s$, and (after possible relabeling), $p_i \sim q_i$ for each i .

Example

Fields are vacuously unique factorization domains.

Example

$\mathbb{Z}$ and $F[x]$ for F a field are unique factorization domains.

Proposition 2.6

Let R be a unique factorization domain, and let $p \in R$. If p is irreducible, then p is prime.

Proof:

Let $p \in R$ be irreducible, and suppose $p \mid ab$ for some $a, b \in R$. Write $ab = pd$ for some $d \in R$. Since R is a UFD, we can factor a, b, d into irreducible elements, say $a = p_1 p_2 \dots p_k$ and $b = q_1 q_2 \dots q_\ell$, and $d = r_1 r_2 \dots r_m$ (taking the convention that we have a unit if these products are empty). Therefore, we have $p r_1 \dots r_m = p_1 \dots p_k q_1 \dots q_\ell$. Since p is irreducible, applying the UFD condition, $p \sim p_i$ for some i or $p \sim q_j$ for some j . Thus $p \mid a$ or $p \mid b$, hence p is prime. ■

Example

Since $\mathbb{Z}$ is a UFD, a prime $p \in \mathbb{Z}$ satisfies Euclid's lemma (i.e. $p \mid ab$ implies $p \mid a$ or $p \mid b$). A similar statement holds if we replace $\mathbb{Z}$ by $F[x]$.

Example

Consider $R = \mathbb{Z}[\sqrt{-5}]$. Then $p = 1 + \sqrt{-5} \in R$ has previously been shown to be irreducible, but not prime. Therefore, R cannot be a unique factorization domain by Proposition 2.6. For example, $6 = (1 + \sqrt{-5})(1 - \sqrt{-5}) = (2)(3)$, so $1 \pm \sqrt{-5} \sim 2$, which is impossible since their norms are 6 and 4 respectively.

Example

However, $R = \mathbb{Z}[\sqrt{-5}]$ satisfies the ascending chain condition on principal ideals. Indeed, if $\langle a_1 \rangle \subseteq \langle a_2 \rangle \subseteq \dots$, we have $a_2 \mid a_1$, $a_3 \mid a_2, \dots$, but taking norms, we have $N(a_1) \geq N(a_2) \geq \dots$, which since these are positive integers, can only strictly decrease finitely many times, implying there must exist $n \in \mathbb{N}$ such that $a_n, a_{n+1}, \dots$ differ by an element of norm 1, i.e. differ by a unit, so $\langle a_n \rangle = \langle a_{n+1} \rangle = \dots$.

Definition

Let R be an integral domain and $a, b \in R$, we say $d \in R$ is a greatest common divisor of a, b , denoted by $\gcd(a, b)$ if it satisfies the following conditions:

- (1) $d \mid a$ and $d \mid b$
- (2) If $e \in R$ with $e \mid a$ and $e \mid b$, then $e \mid d$.

Proposition 2.7

Let R be a unique factorization domain, and let $a, b \neq 0 \in R$. If $p_1, \dots, p_k$ are the non-associated primes dividing a and b , say $a \sim p_1^{\alpha_1} \dots p_k^{\alpha_k}$ and $b \sim p_1^{\beta_1} \dots p_k^{\beta_k}$ with $\alpha_i, \beta_i \in \mathbb{N} \cup \{0\}$. Then $\gcd(a, b) \sim p_1^{\min\{\alpha_1, \beta_1\}} \dots p_k^{\min\{\alpha_k, \beta_k\}}$.

Proof:

Exercise (the proof is basically the same as the integers). ■

Remark

Normal properties of the gcd hold, such as if R is a UFD with $d, a_1, \dots, a_m \in R$, we have $\gcd(da - 1, \dots, da_m) = d \gcd(a_1, \dots, a_m)$.

Theorem 2.8

Let R be an integral domain. The following are equivalent:

- (1) R is a unique factorization domain.
- (2) R satisfies the ascending chain condition on principal ideals and $\gcd(a, b)$ exists for all nonzero $a, b \in R$.
- (3) R satisfies the ascending chain condition on principal ideals and every irreducible element in R is prime.

Proof:

Proving (1) implies (2), let $a, b \neq 0 \in R$. By Proposition 2.7, $\gcd(a, b)$ exists. On the other hand, suppose that there exists a chain $0 \neq \langle a_1 \rangle \subsetneq \langle a_2 \rangle \subsetneq \dots$ in R . Since $\langle a_1 \rangle \neq 0$, we can write $a_1 = p_1^{k_1} \dots p_r^{k_r}$ for non-associated primes p_i and $k_i \in \mathbb{N}$. Since $a_i \mid a_1$ for all i , we have $a_i = p_1^{d_{i,1}} \dots p_r^{d_{i,r}}$ for $0 \leq d_{i,j} \leq k_j$ for all i . Thus there are only finitely many non-associated choices for a_i and so there exists $m \neq n$ with $\langle a_m \rangle = \langle a_n \rangle$, a contradiction. Thus R satisfies the ascending chain condition on principal ideals.

Proving (2) implies (3), we need to show if $p \in R$ is irreducible, then $p \mid ab$ implies $p \mid a$ or $p \mid b$. By (2), let $d = \gcd(a, p)$. Then $d \mid p$. Since p is irreducible, we have $d \sim p$ or $d \sim 1$. If $d \sim p$, then since $d \mid a$, we get $p \mid a$. Thus suppose $d \sim 1$. Then $\gcd(a, p) \sim 1$, so $\gcd(ab, bp) \sim b$. But $p \mid ab$ and $p \mid bp$, so p is a common divisor of ab, bp so $p \mid \gcd(ab, bp) \sim b$ so $p \mid b$.

Proving (3) implies (1), if R satisfies the ascending chain condition on principal ideals, by Theorem 2.4, for $a \in R$ with a nonzero and not a unit, a is a product of irreducible elements. Thus it suffices to show such a factorization is unique up to association. Suppose we have $p_1 \dots p_r \sim q_1 \dots q_s$ with the p_i 's and q_j 's are irreducible. Since p_1 is prime, then $p_1 \mid q_j$ for some j , say without loss of generality q_1 . By Proposition 2.2, we have $p_1 \sim q_1$, implying $p_2 \dots p_r \sim q_2 \dots q_s$. Continue the above process to get $r = s$, and $p_i \sim q_i$ for all i . ■

Definition

An integral domain R is a principal ideal domain (abbreviated PID) if every ideal I of R is principal. That is, every ideal I is of the form $I = \langle a \rangle$ for some $a \in R$.

Example

The two typical PIDs are $\mathbb{Z}$ and $F[x]$ for F a field.

Example

Although all ideals of $\mathbb{Z}_n$ are principal, $\mathbb{Z}_n$ is not a PID unless n is prime, because otherwise it is not an integral domain.

Example

A field F is a PID since its only ideals are $0 = \langle 0 \rangle$ and $F = \langle 1 \rangle$.

Proposition 2.9

Let R be a principal ideal domain, and let $a_1, \dots, a_n$ be nonzero elements of R . Then $\gcd(a_1, \dots, a_n)$ exists and there exist $r_1, \dots, r_n \in R$ such that $\gcd(a_1, \dots, a_n) = r_1 a_1 + \dots + r_n a_n$.

Proof:

We know $A = \langle a_1, \dots, a_n \rangle$ is an ideal of R , so since R is a principal ideal domain, there exists $d \in R$ such that $A = \langle d \rangle$. Thus $d = r_1 a_1 + \dots + r_n a_n$ for some $r_i \in R$. We claim $d \sim \gcd(a_1, \dots, a_n)$. Since $a_i \in \langle a_1, \dots, a_n \rangle = \langle d \rangle$, we see $d \mid a_i$ for each i . d is therefore a common divisor of $a_1, \dots, a_n$. Moreover, if $r \mid a_i$ for each i , then $r \mid (r_1 a_1 + \dots + r_n a_n)$, that is, $r \mid d$. By the definition of the gcd, we see $d \sim \gcd(a_1, \dots, a_n)$. ■

Theorem 2.10

Every principal ideal domain is a unique factorization domain.

Proof:

Let R be a PID. By Theorem 2.8, it suffices to prove R satisfies the ascending chain condition on principal ideal domains, and that $\gcd(a, b)$ exists for all $a, b \in R$, and by Proposition 2.9, the latter holds. Thus, we need to prove the ascending chain condition on R . Suppose $\langle a_1 \rangle \subseteq \langle a_2 \rangle \subseteq \dots$. Then consider $A = \bigcup_{n=1}^{\infty} \langle a_n \rangle$. Then A is an ideal. Since R is a principal ideal domain, we can write $A = \langle a \rangle$ for some $a \in A$. Thus, $a \in \langle a_n \rangle$ for some $n \in \mathbb{N}$ by definition of the union, and hence $\langle a_i \rangle \subseteq A$ for each $i \geq n$, and we already know $a \in \langle a_n \rangle \subseteq \langle a_{n+1} \rangle \subseteq \dots$, thus we see $\langle a \rangle = \langle a_n \rangle = \langle a_{n+1} \rangle = \dots$, so R satisfies the ascending chain condition on principal ideal. It follows that R is a unique factorization domain. ■

Remark

The converse of the above theorem is false, $\mathbb{Z}[x]$ is a counterexample, though proving this is hard (we will do it in section 2.4).

Theorem 2.11

Let R be a principal ideal domain. If $p \in R$ with $p \neq 0$ and not a unit, the following are equivalent:

- (1) p is prime
- (2) $R/\langle p \rangle$ is a field
- (3) $R/\langle p \rangle$ is an integral domain.

Combining this with Theorem 1.4, we see from (2) and (3) that in a principal ideal domain, every nonzero prime ideal is maximal.

Proof:

Proving (1) implies (2), consider an element $a + \langle p \rangle \in R/\langle p \rangle$, and suppose it is nonzero (so $p \nmid a$). Therefore we can consider the ideal $A = \langle a, p \rangle$, which by the fact we are in a PID, is generated by some $d \in R$. But p is prime, and thus irreducible (we are in a PID), so $d \mid p$ implies $d \sim 1$ or $d \sim p$. The latter cannot happen, since $d \sim p$ and $d \mid a$ implies $p \mid a$, a contradiction with our condition above. Therefore,

$d \sim 1$, so $\langle 1 \rangle = \langle a, p \rangle$, so there exists a linear combination $1 = ab + sp$ for $b, s \in R$, implying $ba = 1 - sp$, which mod $\langle p \rangle$ tells us $ba + \langle p \rangle = 1$ in $R/\langle p \rangle$, so $a + \langle p \rangle$ is invertible in $R/\langle p \rangle$, so it is a field (we know it is commutative already by general facts on quotients).

(2) implies (3) is immediate.

Proving (3) implies (1), let $a, b \in R$ and suppose $p \mid ab$ and that $R/\langle p \rangle$ is an integral domain. But then in the quotient ring, $(a + \langle p \rangle)(b + \langle p \rangle) = ab + \langle p \rangle = 0 + \langle p \rangle$ since $p \mid ab$. However, the quotient ring is an integral domain, so either $a = 0 + \langle p \rangle$ or $b = 0 + \langle p \rangle$, that is, either $a \in \langle p \rangle$ or $b \in \langle p \rangle$, so either $p \mid a$ or $p \mid b$. ■

Remark

In a principal ideal domain, the above theorem shows that an ideal is maximal if and only if it is prime (in general only the forward direction holds), and we showed previously in a unique factorization domain that an element is prime if and only if it is irreducible (only the forward direction holds in general).

Example

$\mathbb{Z}[x]$ is not a principal ideal domain. Consider the ideal $A = \{2n + xf(x) : n \in \mathbb{Z}, f(x) \in \mathbb{Z}[x]\}$. Suppose by contradiction that $A = \langle g(x) \rangle$ for some $g(x) \in \mathbb{Z}[x]$. Therefore, we have $g(x) \mid 2 \in A$, so $g(x) \sim 1$ or $g(x) \sim 2$. We cannot have the former, since $1 \notin A$, and we cannot have the latter, since $x \in A$ but $2 \nmid x$. Therefore, we have a contradiction, so A is not principal.

Remark

We have the chain of inclusions

$$\text{ring} \supsetneq \text{comm. ring} \supsetneq \text{integral domain} \supsetneq \text{ACCP} \supsetneq \text{UFD} \supsetneq \text{PID} \supsetneq \text{field}$$

The question is does the strict inclusion between UFDs and PIDs hold? (Yes, $\mathbb{Z}[x]$ is a counterexample). The next section will answer this.

2.4 Gauss' Lemma

Consider the polynomial $2x+4$. It is irreducible in $\mathbb{Q}[x]$, but not irreducible in $\mathbb{Z}[x]$ since $2x+4 = 2(x+2)$, where 2 is no longer a unit. Note that $2 = \gcd(2, 4)$.

Definition

If R is a unique factorization domain and $0 \neq f(x) \in R[x]$, a greatest common divisor of the nonzero coefficients of $f(x)$ is called a content of $f(x)$ and is denoted by $c(f)$. If $c(f) \sim 1$, we say $f(x)$ is a primitive polynomial.

Example

In $\mathbb{Z}[x]$, $c(6 + 10x^2 + 15x^3) \sim \gcd(6, 10, 15) \sim 1$, and $c(6 + 9x^2 + 15x^3) \sim \gcd(6, 9, 15) \sim 3$. Thus $6 + 10x^2 + 15x^3$ is primitive, and $6x + 9x^2 + 15x^3$ is not.

Lemma 2.12

Let R be a unique factorization domain and let $0 \neq f(x) \in R[x]$. Then,

(1) $f(x)$ can be written as $f(x) = c(f)f_1(x)$ where $f_1(x)$ is primitive.

(2) If $0 \neq b \in R$, then $c(bf) \sim bc(f)$

Proof:

Proving (1), for $f(x) = a_mx^m + \dots + a_0 \in R[x]$, let $c = c(f) \sim \gcd(a_0, \dots, a_m)$. Write $a_i = cb_i$ for all i . Then $f(x) = cf_1(x)$ where $f_1(x) = b_mx^m + \dots + b_0$. Then we have $c \sim \gcd(a_0, \dots, a_m) \sim \gcd(cb_0, \dots, cb_m) \sim c \gcd(b_0, \dots, b_m)$. Therefore $c \sim c \gcd(b_0, \dots, b_m)$, so $\gcd(b_0, \dots, b_m) \sim 1$, and hence $f_1(x)$ is primitive.

The proof of (2) is an exercise. ■

Lemma 2.13

Let R be a unique factorization domain and let $\ell(x) \in R[x]$ be irreducible with $\deg \ell \geq 1$. Then $c(\ell) \sim 1$.
Proof:

By the previous lemma, write $\ell(x) = c(\ell)\ell_1(x)$ with $\ell_1(x)$ primitive. Since $\ell(x)$ is irreducible, either $c(\ell)$ or $\ell_1(x)$ is a unit. Since $\deg \ell_1 = \deg \ell \geq 1$, $\ell_1(x)$ is not a unit, thus $c(\ell) \sim 1$. ■

Theorem 2.14 (Gauss' Lemma):

Let R be a unique factorization domain. If $f \neq 0$ and $g \neq 0$ in $R[x]$, then

$$c(fg) \sim c(f)c(g)$$

In particular, the product of primitive polynomials is primitive.

Proof:

Write $f(x) = c(f)f_1(x)$ and $g(x) = c(g)g_1(x)$ by Lemma 2.12 where $f_1(x), g_1(x)$ are primitive. Then by Lemma 2.12(2), we have $c(fg) \sim c(c(f)f_1(x)c(g)g_1(x)) \sim c(f)c(g)c(f_1(x)g_1(x))$. Thus it suffices to prove the theorem for primitive polynomials: that $f_1(x)g_1(x)$ is primitive. Suppose for the sake of contradiction that $f_1(x)g_1(x)$ is not primitive. Since R is a unique factorization domain, there exists a prime p dividing each coefficient of $f_1(x)g_1(x)$. Write $f_1(x) = a_0 + \cdots + a_mx^m$ and $g_1(x) = b_0 + \cdots + b_nx^n$. Since $f_1(x), g_1(x)$ are primitive, p does not divide every a_i nor every b_j . Thus there exists $k, s \in \mathbb{N} \cup \{0\}$ s.t. $p \nmid a_k$ but $p \mid a_i$ for $0 \leq i < k$ and $p \nmid b_s$ but $p \mid b_j$ for $0 \leq j < s$. The coefficient of x^{k+s} in $f(x)g(x)$ is $c_{k+s} = \sum_{i+j=k+s} a_i b_j$. Because of the divisibility conditions, p divides all $a_i b_j$ for $i+j = k+s$ except $a_k b_s$. It follows that $p \nmid c_{k+s}$, a contradiction. Thus $f_1(x)g_1(x)$ is primitive. ■

Theorem 2.15

Let R be a UFD whose field of fractions is F . Regard $R \subseteq F$ as a subring of F in the usual way. If $\ell(x) \in R[x]$ is irreducible in $R[x]$, then $\ell(x)$ is irreducible in $F[x]$.

Proof:

Let $\ell(x) \in R[x]$ be irreducible. Suppose $\ell(x) = g(x)h(x)$ over $F[x]$. If a and b are the product of the denominators of the coefficients of $g(x)$ and $h(x)$, then $g_1(x) = ag(x) \in R[x]$ and $h_1(x) = bh(x) \in R[x]$. Note that $ab\ell(x) = g_1(x)h_1(x)$ is a factorization of $\ell(x)$ in $R[x]$. Since $\ell(x)$ is irreducible in $R[x]$, by Lemma 2.13, $c(\ell) \sim 1$, and by Gauss' Lemma, $ab \sim abc(\ell(x)) \sim c(ab\ell(x)) \sim c(g_1(x)h_1(x)) \sim c(g_1(x))c(h_1(x))$ (denote this as $(\star)$). Now write $g_1(x) = c(g_1)g_2(x)$ and $h_1(x) = c(h_1)h_2(x)$, where $g_2(x)h_2(x)$ are primitive in $R[x]$. Then $ab\ell(x) = g_1(x)h_1(x) = c(g_1)c(h_1)g_2(x)h_2(x)$. By $(\star)$, we have $\ell(x) \sim g_2(x)h_2(x)$ in $R[x]$, so by irreducibility, either $g_2(x)$ or $h_2(x)$ is a unit in $R[x]$. Suppose $g_2(x)$ is a unit in $R[x]$. Then $ag(x) = g_1(x) = c(g_1)g_2(x)$, so $g(x) = a^{-1}c(g_1)g_2(x)$ with $g_2(x) \sim 1$, which is a unit in $F[x]$. Similarly, if $h_2(x) \sim 1$, then $h(x)$ is a unit in $F[x]$. Thus $\ell(x) = g(x)h(x)$ in $F[x]$ implies that either $g(x)$ or $h(x)$ is a unit in $F[x]$, so $\ell(x)$ is irreducible in $F[x]$. ■

Remark

We see from the above proof that if $f(x) \in R[x]$ admits a factorization in $F[x]$ as $f(x) = g(x)h(x)$, then by Gauss' Lemma, there exist $\tilde{g}(x), \tilde{h}(x)$ in $R[x]$. For example, $2x^2 + 7x + 3 = (x + \frac{1}{2})(2x + 6) = (2x + 1)(x + 3)$.

Remark

The converse of Theorem 2.15 is false, for example $2x + 4$ is irreducible in $\mathbb{Q}[x]$, but $2x + 4 = 2(x + 2)$ is reducible in $\mathbb{Z}[x]$. Note that $2 = c(2x + 4)$. This is the only way the converse can fail:

Proposition 2.16

Let R be a unique factorization domain with field of fractions F . Regard $R \subseteq F$ as a subring of F in the usual way. Let $f(x) \in R[x]$ with $\deg f \geq 1$. Then $f(x)$ is irreducible in $R[x]$ if and only if $f(x)$ is primitive in $R[x]$ and irreducible in $F[x]$.

Proof:

The forward direction is Lemma 2.13 combined with Theorem 2.15. Thus in the reverse direction, suppose

$f(x)$ is primitive in $R[x]$ and irreducible in $F[x]$, but reducible in $R[x]$. Then a nontrivial factorization of $f(x)$ in $R[x]$ must be of the form $f(x) = dg(x)$ with $d \in R$ and $d \not\sim 1$, $\deg g(x) \geq 1$, where we know that $\deg d = 0$ since otherwise we would have $f(x) = d(x)g(x)$ which would also be a nontrivial factorization in $F[x]$, contradicting irreducibility. Since $d \mid f(x)$, we have that d must divide each coefficient of $f(x)$, so $d \mid c(f)$ by properties of the GCD, contradicting primitivity since $d \not\sim 1$. A contradiction, so $f(x)$ is irreducible in $R[x]$. ■

Theorem 2.17

Let R be a unique factorization domain. Then the polynomial ring $R[x]$ is also a unique factorization domain.

Proof:

Since R satisfying the ascending chain condition on principal ideals implies $R[x]$ does, by Theorem 2.8, it suffices to prove every irreducible element in $R[x]$ is prime. Let $\ell(x)$ be irreducible in $R[x]$ with $\ell(x) \mid f(x)g(x)$ for $f(x), g(x) \in R[x]$. If $\deg \ell = 0$, then taking contents, we have $\ell = c(\ell)$ and thus $\ell \mid c(fg) = c(f)c(g)$. Since ℓ is prime in R (since ℓ is irreducible and R is a UFD) then $\ell \mid c(f)$ or $\ell \mid c(g)$, corresponding to $\ell(x) \mid f(x)$ or $\ell \mid g(x)$ respectively. Thus we may assume $\deg \ell \geq 1$, where $c(\ell) \sim 1$ by Lemma 2.13. Then, taking the original divisibility, there exists $h(x) \in R[x]$ such that $\ell(x)h(x) = f(x)g(x)$, which taking contents with Lemma 2.12, implies $\ell(x)c(h)h_1(x) = c(f)c(g)f_1(x)g_1(x)$ for $f_1(x), g_1(x), h_1(x) \in R[x]$ primitive. Since $c(\ell) \sim 1$, we have $c(h) \sim c(f)c(g)$, and so we have $h_1(x)\ell(x) \sim f_1(x)g_1(x)$, so $\ell(x) \mid f_1(x)g_1(x)$, and thus it suffices to prove the theorem for the case of $f_1(x), g_1(x)$ primitive. Let F denote the field of fractions of R and regard R as a subring of F in the usual way. Then we have $\ell(x) \mid f_1(x)g_1(x)$ in $F[x]$. Since $\ell(x) \in R[x]$ is irreducible, it is irreducible in $F[x]$, and thus by Euclid's lemma for $F[x]$ (we are in a field), we have $\ell(x) \mid f_1(x)$ or $\ell(x) \mid g_1(x)$, without loss of generality suppose it is $f_1(x)$. Write $f_1(x) = \ell(x)k(x)$ for $k(x) \in F[x]$. If d is the product of all denominators of coefficients in $k(x)$, then $k_0(x) := dk(x) \in R[x]$, and we have $df_1(x) = k_0(x)\ell(x)$. Since $f_1(x)$ is primitive and $c(\ell) \sim 1$, by Gauss' Lemma, we have $d \sim c(df_1) \sim c(k_0\ell) \sim c(k_0)c(\ell) \sim c(k_0)$. Thus writing $k_0(x) = c(k_0)k_1(x)$ for $k_1(x) \in R[x]$ primitive, then $df_1(x) = k_0(x)\ell(x) = c(k_0)k_1(x)\ell(x)$, and so since $d \sim c(k_0)$, we have $f_1(x) \sim k_1(x)\ell(x)$, and so $\ell(x) \mid f_1(x)$ over $R[x]$. Therefore, $\ell(x)$ is prime in $R[x]$, and so $R[x]$ is a unique factorization domain. ■

Corollary 2.18

If R is a unique factorization domain, then for all $n \in \mathbb{N}$, $R[x_1, \dots, x_n]$ is a unique factorization domain. ■

Corollary 2.19

$\mathbb{Z}[x]$ and $\mathbb{Z}[x_1, \dots, x_n]$ are unique factorization domains. ■

Remark

Since $\mathbb{Z}[x]$ is not a principal ideal domain, we have that PIDs $\subsetneq$ UFDs.

Theorem 2.20 (Eisenstein's Criterion for UFDs)

Let R be a unique factorization domain and let F be its field of fractions. Let $h(x) = c_n x^n + \dots + c_1 x + c_0 \in R[x]$ with $n \geq 1, c_n \neq 0$. Let $\ell \in R$ be an irreducible element. If $\ell \nmid c_n, \ell \mid c_i$ for all $0 \leq i < n$, and $\ell^2 \nmid c_0$, then $h(x)$ is irreducible in $F[x]$.

Proof:

We prove by contradiction. If $h(x)$ is reducible in $F[x]$, by Gauss' Lemma for unique factorization domains, there exist $s(x), r(x) \in R[x]$ of positive degree such that $h(x) = s(x)r(x)$. Write $s(x) = a_0 + \dots + a_m x^m$, and $r(x) = b_0 + \dots + b_k x^k$ where $1 \leq m, k < n$. Since $h(x) = s(x)r(x)$, $c_0 = a_0 b_0$, $c_1 = a_0 b_1 + a_1 b_0$, $c_2 = a_0 b_2 + a_1 b_1 + a_2 b_0$, and so on. Consider the constant term. Since $\ell \mid c_0$, we have $\ell \mid a_0 b_0$, and since ℓ is irreducible, we have $\ell \mid a_0$ or $\ell \mid b_0$. Without loss of generality suppose $\ell \mid a_0$. Since $\ell^2 \nmid c_0$, we have $\ell \nmid b_0$. Similarly considering the coefficient of x , since $\ell \mid c_1$, we have $\ell \mid (a_0 b_1 + a_1 b_0)$, and thus $\ell \mid a_1 b_0$, which since ℓ is irreducible and $\ell \nmid b_0$, we have $\ell \mid a_1$. This process repeats, and we have $\ell \mid a_i$ for all $i \in \{0, \dots, m-1\}$. But $\ell \nmid a_n$ since $\ell \nmid c_n$. Consider the reduction $\bar{h}(x) = \bar{s}(x)\bar{r}(x)$ in

$(R/\langle \ell \rangle)[x]$, where the bars indicate the mod $\langle \ell \rangle$ reductions of the coefficients. By the assumption of the divisibility of the coefficients of $h(x)$, we have $\bar{h}(x) = \bar{c}_n x^n$. However, since $\bar{s}(x) = \bar{a}_m x^m$ (since $\ell \nmid a_m$) and $\ell \nmid b_0$, $\bar{s}(x)\bar{r}(x)$ contains the nonzero term $\bar{a}_m \bar{b}_0 x^m$, and all other terms with a $\bar{a}_i$ for $i \in \{0, \dots, m-1\}$ will vanish (so there is no other term of degree m , so nothing will cancel), with $m < n$. Therefore, we have calculated something different on both sides for $\bar{h}(x) = \bar{c}_n x^n$, and $\bar{s}(x)\bar{r}(x) = \bar{c}_n x^n + \bar{a}_m \bar{b}_0 x^m$, a contradiction, so $h(x)$ must be not reducible. ■

Remark

Since $\mathbb{Z}$ is a unique factorization domain, Theorem 2.20 holds with $R = \mathbb{Z}$, $F = \mathbb{Q}$, and ℓ a prime number.

Example

The polynomial $2x^7 + 3x^4 + 6x^2 + 12$ is irreducible in $\mathbb{Q}[x]$ with the prime $p = 3$ using Eisenstein's criterion.

Example

Let p a prime, and let $\zeta_p = e^{2\pi i/p}$ be a p th root of 1. It is a root of the p th cyclotomic polynomial $\Phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \dots + 1$. Eisenstein's criterion does not imply the irreducibility of $\Phi_p(x)$ immediately. However,

$$\Phi_p(x+1) = \frac{(x+1)^{p-1} - 1}{x} = x^{p-1} + \binom{p}{1}x^{p-2} + \binom{p}{3}x^{p-3} + \dots + \binom{p}{p-2}x + \binom{p}{p-1}$$

is a polynomial in $\mathbb{Z}[x]$. Since p is a prime, $p \nmid 1$, and $p \mid \binom{p}{i}$ for $1 \leq i < p$, and $p^2 \nmid p = \binom{p}{p-1}$. Thus by Eisenstein's criterion, $\Phi_p(x+1)$ is irreducible in $\mathbb{Q}[x]$. This implies $\Phi_p(x)$ is also irreducible in $\mathbb{Q}[x]$ (note that the map $x \mapsto x+1$ is a ring isomorphism of $\mathbb{Q}[x]$, and thus does not change irreducibility of a polynomial). Moreover, by Proposition 2.16, since $\Phi_p(x)$ is primitive in $\mathbb{Z}[x]$, it is also irreducible in $\mathbb{Z}[x]$.

Chapter 3: Field Extensions

3.1 Degree of Extensions

Definition

If E is a field containing another field F , we say E is a field extension of F , denoted by E/F .

Remark

It might feel confusing to use the same notation as the quotient, but quotients of fields aren't very meaningful since the only ideals of a field are the whole field and the zero ideal, and so the only quotients are the trivial ring and the field unchanged, so we're not really doubling up here, and it will be safe to realize that in field theory, the slash is always a field extension.

If E/F is a field extension, we can view E as a vector space over F in the following way for $c \in F, e_1, e_2 \in E$:

- (1) $\underbrace{e_1 + e_2}_{+ \text{ in vector space}} := \underbrace{e_1 + e_2}_{+ \text{ in } E}$
- (2) $\underbrace{ce_1}_{\text{scalar mult. in vector space}} := \underbrace{ce_1}_{\text{mult. in } E}$

Definition

Given a field extension E/F viewed as a vector space E over a field F , its dimension as a vector space is called the degree of E over F and denoted by $[E : F]$. If $[E : F]$ is finite, then we say E/F is a finite extension, otherwise, we say it is an infinite extension.

Example

$[\mathbb{C} : \mathbb{R}] = 2$ since $\mathbb{C} \cong \mathbb{R} + i\mathbb{R}$.

Example

Let F be a field and consider the polynomial ring $F[x]$. Its field of fractions $F(x)$ is the set of rational functions with coefficients in F . Then $[F(x) : F]$ is infinite since $\{1, x, x^2, \dots\}$ is an infinite linearly independent set over F .

Theorem 3.1

If E, K, F are fields, and E/K and K/F are finite field extensions, then E/F is a finite field extension and additionally $[E : F] = [E : K][K : F]$. In particular, if K is an intermediate field of a finite extension E/F , then $[K : F] \mid [E : F]$.

Proof:

Suppose $[E : K] = m$ and $[K : F] = n$. Let $\{a_1, \dots, a_m\}$ be a basis of E/K and $\{b_1, \dots, b_n\}$ be a basis of K/F . It suffices to show $B := \{a_i b_j : 1 \leq i \leq m, 1 \leq j \leq n\}$ is a basis of E/F . We first prove that B spans. For $e \in E$, we have that there exist $k_1, \dots, k_m \in K$ such that $e = \sum_{i=1}^m k_i a_i$. For each $k_i \in K$, we have that there exist $c_{i,1}, \dots, c_{i,n} \in F$ such that $k_i = \sum_{j=1}^n c_{i,j} b_j$. Therefore $e = \sum_{i=1}^m \sum_{j=1}^n c_{i,j} b_j a_i$ and so B spans. We now show B is linearly independent. Suppose that $\sum_{i=1}^m \sum_{j=1}^n c_{i,j} b_j a_i = 0$. Now $\sum_{j=1}^n c_{i,j} b_j$ is an element of K , and thus since $a_1, \dots, a_m$ are linearly independent over K , we must have for each i that $0 = \sum_{j=1}^n c_{i,j} b_j$, which is an F -linear combination, but now by linear independence of $b_1, \dots, b_n$ over F , we have that $c_{i,j} = 0$ for all j , which is what we needed to show to show linear independence. Therefore B spans and is linearly independent, so B is a basis of E/F and we have $[E : F] = [E : K][K : F]$. ■

3.2 Algebraic and Transcendental Extensions

Definition

Let E/F be a field extension and let $\alpha \in E$. We say α is algebraic over F if there exists $f(x) \in F[x] \setminus \{0\}$ with $f(\alpha) = 0$. If α is not algebraic over F , we say it is transcendental over F .

Example

Any element $a \in \mathbb{Q}$ and $\sqrt{2}$ are algebraic over $\mathbb{Q}$ (the former a root of polynomial $x - a$, the latter a root of polynomial $x^2 - 2$).

Example

$\alpha = \sqrt{2} + \sqrt{3}$ is algebraic over $\mathbb{Q}$. To see this, write $\alpha - \sqrt{2} = \sqrt{3}$, square both sides to get $\alpha^2 - 2\sqrt{2}\alpha + 2 = 3$, so $\alpha^2 - 1 = 2\sqrt{2}\alpha$, so $\alpha^4 - 2\alpha^2 + 1 = 8\alpha^2$, so $\alpha^4 - 10\alpha^2 + 1 = 0$, so α is a root of $x^4 - 10x^2 + 1$.

Definition

Let E/F be a field extension and $\alpha \in E$, and let $F[\alpha]$ denote the smallest subring of E containing F and α , and $F(\alpha)$ denote the smallest subfield of E containing F and α . Similarly define $F[\alpha, \beta]$ and $F(\alpha, \beta)$, and so on for $\alpha, \beta \in E$.

Definition

Let E/F be a field extension. If there exists $\alpha \in E$ such that $E = F(\alpha)$, we say E is a simple extension of F .

Definition

Let R and R_1 be two rings which contain a field F . A ring homomorphism $\psi : R \rightarrow R_1$ is called an F -homomorphism if $\psi|_F$ is the identity map on F .

Theorem 3.2

Let E/F be a field extension and let $\alpha \in E$. If α is transcendental over F , then $F[\alpha] \cong F[x]$ and $F(\alpha) \cong F(x)$, and in particular $F[\alpha] \neq F(\alpha)$.

Proof:

Let $\psi : F(x) \rightarrow F(\alpha)$ be the F -homomorphism uniquely determined by $\psi(x) = \alpha$. Thus for $f(x), g(x) \in F[x]$ with $g(x) \neq 0$, $\psi(\frac{f(x)}{g(x)}) = \frac{f(\alpha)}{g(\alpha)} \in F(\alpha)$, where we see this map is well-defined since α is transcendental, and thus not a root of any polynomial in $F[x]$, and thus $g(\alpha) \neq 0$. Now, since $F(x)$ is a field and $\ker \psi$ is an ideal of $F(x)$, we have $\ker \psi = 0$ (since ψ is nontrivial, for instance $\psi(x) = \alpha \neq 0$) and so we see ψ is injective. Moreover, since $\text{im } \psi$ contains a field generated by F and α , by minimality, $F(\alpha) \subseteq \text{im } \psi$, so $\text{im } \psi = F(\alpha)$ and so ψ is surjective. Thus ψ is an isomorphism, so $F(\alpha) \cong F(x)$, and restricting its domain to $F[\alpha]$ further shows that $F[\alpha] \cong F[x]$. ■

Theorem 3.3

Let E/F be a field extension, and let $\alpha \in E$. If α is algebraic over F , there exists a unique monic irreducible polynomial $p(x) \in F[x]$ such that there exists a F -isomorphism $\psi : F[x]/\langle p(x) \rangle \rightarrow F[\alpha]$ with $\psi(x) = \alpha$, from which we conclude $F[\alpha] = F(\alpha)$.

Proof:

Consider the F -homomorphism $\psi : F[x] \rightarrow F[\alpha]$ uniquely determined by $\psi(x) = \alpha$. Thus for all $f(x) \in F[x]$, we have $\psi(f) = f(\alpha) \in F[\alpha]$. Since $F[x]$ is a ring, $\text{im } \psi$ contains a ring generated by F and α , i.e. $F[\alpha] \subseteq \text{im } \psi$, so $\text{im } \psi = F[\alpha]$. Now set $I = \ker \psi = \{f(x) \in F[x], f(\alpha) = 0\}$. Since α is algebraic, $I \neq \{0\}$. So by the first isomorphism theorem, $F[x]/I \cong \text{im } \psi = F[\alpha]$, which is a subring of the field $F(\alpha)$. Thus $F[x]/I$ is an integral domain, so I is a prime ideal. It follows that $I = \langle p(x) \rangle$, where $p(x)$ is irreducible, where we can assert $p(x)$ is unique by making it monic. It follows that $F[x]/\langle p(x) \rangle \cong F[\alpha]$, where since $F[x]$ is a principal ideal domain, $\langle p(x) \rangle$ is maximal, so $F[\alpha]$ is a field. Therefore, by minimality of $F(\alpha)$ (it's the smallest field containing F and α), we get $F[\alpha] = F(\alpha)$. ■

Definition

Let E/F be a field extension. If $\alpha \in E$ is algebraic over F , the unique monic irreducible polynomial in the above theorem is called the minimal polynomial of α over F . From the proof, if $f(x) \in F[x]$ with $f(\alpha) = 0$, then $p(x) \mid f(x)$.

Theorem 3.4

Let E/F be a field extension and let $\alpha \in E$.

- (1) α is transcendental over F if and only if $[F(\alpha) : F]$ is infinite.
- (2) α is algebraic over F if and only if $[F(\alpha) : F] < \infty$. Moreover, if $p(x)$ is the minimal polynomial of α over F , we have $[F(\alpha) : F] = \deg p$, and $\{1, \alpha, \alpha^2, \dots, \alpha^{\deg p - 1}\}$ is a basis of $F(\alpha)/F$.

Proof:

By taking contrapositives, it suffices to prove only the forward directions in (1) and (2). For (1), by Theorem 3.2, if α is transcendental over F , $F(\alpha) \cong F(x)$. But in $F(x)$, the elements $\{1, x, x^2, \dots\}$ are linearly independent over F , thus $[F(\alpha) : F]$ is infinite. Proving (2), suppose α is algebraic over F . By Theorem 3.3, $F(\alpha) \cong F[x]/\langle p(x) \rangle \cong \{r(x) \in F[x], \deg r < \deg p\}$. Thus $\{1, x, \dots, x^{\deg p - 1}\}$ forms a basis of $F[x]/\langle p(x) \rangle$ over F . Since our isomorphism $F(\alpha) \cong F[x]/\langle p(x) \rangle$ sends x to α , it follows that $\{1, \dots, \alpha^{\deg p - 1}\}$ is a basis of $F[\alpha]$ over F , so $[F(\alpha) : F] = \deg p$. ■

Example

We saw in Chapter 2 that ζ_p is a root of the p th cyclotomic polynomial $\Phi_p(x)$, which is irreducible. Since $\Phi_p(x)$ is monic, it must be the minimal polynomial of ζ_p over $\mathbb{Q}$ and $[\mathbb{Q}(\zeta_p) : \mathbb{Q}] = p - 1$. The extension $\mathbb{Q}(\zeta_p)/\mathbb{Q}$ is called the p th cyclotomic extension of $\mathbb{Q}$.

Example

Let $\alpha = \sqrt{2} + \sqrt{3}$. We have seen before that α is a root of the polynomial $x^4 - 10x^2 + 1$. Thus

$[\mathbb{Q}(\alpha) : \mathbb{Q}] \leq 4$. One can show $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Since $\sqrt{2}$ is a root of $x^2 - 2$ which is irreducible, $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$. Since $\sqrt{3}$ and $\sqrt{2}$ are linearly independent over $\mathbb{Q}$, $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$ and thus $\alpha \notin \mathbb{Q}(\sqrt{2})$. Hence $[\mathbb{Q}(\alpha) : \mathbb{Q}(\sqrt{2})] \geq 2$. Thus $4 \geq [\mathbb{Q}(\alpha) : \mathbb{Q}] = [\mathbb{Q}(\alpha) : \mathbb{Q}(\sqrt{2})][\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] \geq 2 \cdot 2 = 4$. Hence $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4$ and $x^4 - 10x^2 + 1$ is the minimal polynomial of α over $\mathbb{Q}$, and thus it must be irreducible.

Theorem 3.5

Let E/F be a field extension. If E/F is a finite extension, then there exist $\alpha_1, \dots, \alpha_n \in E$ such that

$$F \subsetneq F(\alpha_1) \subsetneq F(\alpha_1, \alpha_2) \subsetneq \cdots \subsetneq F(\alpha_1, \dots, \alpha_n) = E$$

Proof:

We prove this by induction on $[E : F]$. If $[E : F] = 1$, then $E = F$ and we are done. Suppose $[E : F] > 1$ and the statement holds for all field extensions E_1/F_1 with $[E_1 : F_1] < [E : F]$. Let $\alpha_1 \in E \setminus F$. By Theorem 3.1, $[E : F] = [E : F(\alpha_1)][F(\alpha_1) : F]$. Since $[F(\alpha_1) : F] > 1$, we have $[E : F(\alpha_1)] < [E : F]$. By the induction hypothesis, there exist $\alpha_2, \dots, \alpha_n$ such that

$$F(\alpha_1) \subsetneq F(\alpha_1)(\alpha_2) \subsetneq \cdots \subsetneq F(\alpha_1)(\alpha_2, \dots, \alpha_n) = E$$

and the result holds by induction. ■

Remark

By Theorem 3.5, to understand a finite extension, it suffices to understand a finite simple extension, since we can decompose any finite extension into a finite chain of simple extensions.

Definition

A field extension E/F is an algebraic extension if every $\alpha \in E$ is algebraic over F . Otherwise it is a transcendental extension.

Theorem 3.6

Let E/F be a field extension. If $[E : F] < \infty$, then E/F is algebraic.

Proof:

Suppose $[E : F] = n$. For $\alpha \in E$, the elements $\{1, \alpha^2, \dots, \alpha^n\}$ are linearly dependent over F , thus there exist $c_i \in F$ for $0 \leq i \leq n$ not all 0, such that $\sum_{i=0}^n c_i \alpha^i = 0$, so α is a root of the polynomial $\sum_{i=0}^n c_i x^i \in F[x]$, thus α is algebraic over F . ■

Theorem 3.7

Let E/F be a field extension. Define $L = \{\alpha \in E : [F(\alpha) : F] < \infty\}$. Then L is a field with $F \subseteq L \subseteq E$.

Proof:

We need to show $\alpha \pm \beta$, $\alpha\beta$ and α/β with $\beta \neq 0$ are elements of L . Consider $F(\alpha, \beta)/F$. We claim this is a finite extension, indeed, let $p_1 \in F[x]$ be the minimal polynomial of β over F , and let $p_2(x) \in F(\alpha)[x]$ be the minimal polynomial of β over $F(\alpha)$ (in particular β is algebraic over $F(\alpha)$ since it is algebraic over F). Then $p_2 \mid p_1$ because $p_1 \in F(\alpha)[x]$ and has β as a root, so we see $\deg p_2 \leq \deg p_1$, and so $[F(\alpha, \beta) : F(\alpha)] = \deg p_2 \leq \deg p_1 = [F(\beta) : F]$, and so $[F(\alpha, \beta) : F] = [F(\alpha, \beta) : F(\alpha)][F(\alpha) : F] \leq [F(\beta) : F][F(\alpha) : F]$, which is finite by Theorem 3.4 since α and β are algebraic. Thus, since $F(\alpha \pm \beta), F(\alpha\beta), F(\alpha/\beta) \subseteq F(\alpha, \beta)$, they are finite extensions of F , so $\alpha \pm \beta, \alpha\beta, \alpha/\beta \in L$. ■

Definition

Let E/F be a field extension. The field $L = \{\alpha \in E : [F(\alpha) : F] < \infty\}$ is called the algebraic closure of F in E .

Example

By the fundamental theorem of algebra, $\mathbb{C}$ is the algebraic closure of $\mathbb{R}$ in $\mathbb{C}$.

Example

What about a converse to Theorem 3.6? Let $\overline{\mathbb{Q}}$ be the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$. Since for every prime p , a p th root of unity ζ_p is in $\overline{\mathbb{Q}}$ (as a root of the p th cyclotomic polynomial), we have $[\overline{\mathbb{Q}} : \mathbb{Q}] \geq [\mathbb{Q}(\zeta_p) : \mathbb{Q}] = p - 1$. But $p \rightarrow \infty$, so $[\overline{\mathbb{Q}} : \mathbb{Q}]$ is infinite, hence the converse to Theorem 3.6 is false.

Chapter 4: Splitting Fields

Definition

Let E/F be a field extension. We say $f(x) \in F[x]$ splits over E if E contains all roots of $f(x)$ ($f(x)$ is a product of linear factors in $E[x]$).

Definition

Let E/F be a field extension, $f(x) \in F[x]$ and $F \subseteq G \subseteq E$. If $f(x)$ splits over G and there is no proper subfield of G that $f(x)$ splits over, then we say G is a splitting field of $f(x)$ in E .

Theorem 4.1

Let F be a field and let $p(x) \in F[x]$ be irreducible. The quotient ring $F[x]/\langle p(x) \rangle$ is a field containing F and a root of $p(x)$.

Proof:

Since $p(x)$ is irreducible in the PID $F[x]$, $\langle p(x) \rangle$ is prime and thus maximal, so $E = F[x]/\langle p(x) \rangle$ is a field. Consider the map $g : F \rightarrow E$ by $a \mapsto a + I$ for $a \in F$. Since g is nonzero and F is a field, g is injective, so we can identify F with $g(F)$ to view F as a subfield of E . Let $b = x + I \in E$. We claim b is a root of $p(x)$ in E . Write $p(x) = (a_0 + I) + \cdots + (a_n + I)x^n \in E[x]$ for $a_i \in F$. Then

$$p(b) = (a_0 + I) + (a_1 + I)(x + I) + \cdots + (a_n + I)(x^n + I) = (a_0 + a_1 + \cdots + a_n x^n) + I = p(x) + I = 0 + I$$

so b is a root of $p(x)$. ■

Theorem 4.2 (Kronecker)

Let F be a field and let $f(x) \in F[x]$. There exists a field E containing F such that $f(x)$ splits over E .

Proof:

We prove this by induction on the degree of $f(x)$. The base case degree 1 is clear, since all degree 1 polynomials in $F[x]$ already have all of their roots in F .

Assume inductively that every polynomial of degree less than $n := \deg f$ can be split in some field E containing F . Write $f(x) = p(x)h(x)$ where $p(x) \in F[x]$ is irreducible. By Theorem 4.1, there exists a field K containing F and a root of $p(x)$. Call the root α . Thus $p(x) = (x - \alpha)q(x)$, $f(x) = (x - \alpha)q(x)h(x)$ where $q(x) \in K[x]$. Since $\deg(hq) < n$, we may apply the inductive hypothesis to get a field E containing K (containing F) in which $h(x)q(x)$ splits, where we see this means that $f(x)$ too splits over E in this case. ■

Theorem 4.3

Let F be a field and let $f(x) \in F[x]$. Then it has a splitting field that is a finite extension of F .

Proof:

For $f(x) \in F[x]$, we can find a field extension E/F in which $f(x)$ splits by Kronecker's theorem. Let $\alpha_1, \dots, \alpha_n$ be the roots of $f(x)$ in E . Consider $F(\alpha_1, \dots, \alpha_n)$. This is the smallest subfield of E containing all roots of $f(x)$. So, $f(x)$ does not split over any proper subfield of $F(\alpha_1, \dots, \alpha_n)$, so $F(\alpha_1, \dots, \alpha_n)$ is the splitting field of $f(x)$ in E . Since all the α_i are algebraic, the extension is finite. ■

Remark

If $f(x)$ splits in E with roots $\alpha_1, \dots, \alpha_n \in E$, then $F(\alpha_1, \dots, \alpha_n)$ is the splitting field of $f(x)$ in E .

4.2 Uniqueness of Splitting Fields

We have seen that for the field extension E/F , $F(\alpha_1, \dots, \alpha_n)$ is the splitting field of $f(x) \in F[x]$ in E and it is unique in E . The question is if we change E/F to a different extension, say E_1/F , what is the relation between the splitting fields of $f(x)$ in E and E_1 ?

Definition

Let $\phi : R \rightarrow R_1$ be a ring homomorphism and $\Phi : R[x] \rightarrow R_1[x]$ be the ring homomorphism uniquely determined by $\Phi|_R = \phi$ and $\Phi(x) = x$. In this case, we say Φ extends ϕ . More generally, if $R \subseteq S$, $R_1 \subseteq S_1$ are subrings, and $\Phi : S \rightarrow S_1$ is a ring homomorphism, with $\Phi|_R = \phi$, we say Φ extends ϕ .

Theorem 4.4

Let $\phi : F \rightarrow F_1$ be an isomorphism of fields and $f(x) \in F[x]$. Let $\Phi : F[x] \rightarrow F_1[x]$ be the unique ring isomorphism that extends ϕ . Let $f_1(x) = \Phi(f(x))$ and E/F and E_1/F_1 be splitting fields of $f(x)$ and $f_1(x)$ respectively. Then there exists an isomorphism of fields $\psi : E \rightarrow E_1$ which extends ϕ .

Proof:

We prove this by induction on $[E : F]$. If $[E : F] = 1$, then $E = F$, so $f(x)$ is a product of linear factors in $F[x]$, and thus $f_1(x)$ is a product of linear factors in $F_1[x]$ by the isomorphism ϕ . Thus $E_1 = F_1$, and taking $\psi = \phi$ gives the result in this case. Now suppose $[E : F] > 1$ and the statement is true for all field extensions $\tilde{E}/\tilde{F}$ with $[\tilde{E} : \tilde{F}] < [E : F]$. Let $p(x) \in F[x]$ be an irreducible factor of $f(x)$ with $\deg p \geq 2$, and let $p_1(x) = \Phi(p(x))$, where such a $p(x)$ exists, since if all irreducible factors of $f(x)$ are of degree 1, then $f(x)$ splits in F and we have $E = F$, which is impossible by our condition on the degree. Let $\alpha \in E$ and $\alpha_1 \in E_1$ be roots of $p(x)$ and $p_1(x)$. By Theorem 3.3, we have an F -isomorphism $F(\alpha) \cong F[x]/\langle p(x) \rangle$, $\alpha \mapsto x + \langle p(x) \rangle$ and an F_1 -isomorphism $F_1(\alpha_1) \cong F_1[x]/\langle p_1(x) \rangle$, $\alpha_1 \mapsto x + \langle p_1(x) \rangle$. Since $p_1(x) = \Phi(p(x))$, this gives a field isomorphism (since Φ is an isomorphism), $\tilde{\Phi} : F[x]/\langle p(x) \rangle \rightarrow F_1[x]/\langle p_1(x) \rangle$, $x + \langle p(x) \rangle \mapsto x + \langle p_1(x) \rangle$ which extends ϕ . Since $F[x]/\langle p(x) \rangle \cong F(\alpha)$, and similarly for $F_1(\alpha_1)$, this gives a field isomorphism $\tilde{\phi} : F(\alpha) \rightarrow F_1(\alpha_1)$ determined by $\alpha \mapsto \alpha_1$ which extends ϕ . Note that since $\deg p \geq 2$, $[E : F(\alpha)] < [E : F]$. Since E (respectively E_1) is the splitting field of $f(x) \in F(\alpha)[x]$ (respectively $f_1(x) \in F_1(\alpha_1)[x]$) over $F(\alpha)$ (respectively $F_1(\alpha_1)$), by induction, there exists $\psi : E \rightarrow E_1$ which extends $\tilde{\phi}$, thus ψ also extends ϕ . ■

Corollary 4.5

Any two splitting fields of $f(x) \in F[x]$ over F are F -isomorphic. Thus we can now talk about “the” splitting field of $f(x)$ over F .

Proof:

Take $\phi : F \rightarrow F$ as the identity, and apply the above theorem. ■

4.3 Degrees of Splitting Fields

Theorem 4.6

Let F be a field and $f(x) \in F[x]$ with $\deg f = n \geq 1$. If E/F is the splitting field of $f(x)$, then $[E : F] \mid n!$

Proof:

We prove this theorem by induction on n . If $n = 1$, then $E = F$ and we have $1 \mid 1$ and the statement holds. Now suppose $\deg f > 1$ and the statement holds for all $g(x)$ with $\deg g < \deg f$. Now suppose $f(x) \in F[x]$ is irreducible with $\alpha \in E$ a root of $f(x)$ by Theorem 3.3. Then $F(\alpha) \cong F[x]/\langle f(x) \rangle$ and $[F(\alpha) : F] = \deg f = n$. Write $f(x) = (x - \alpha)g(x) \in F(\alpha)[x]$ with $g(x) \in F(\alpha)[x]$. By minimality of the splitting field, we must have that $E/F(\alpha)$ is the splitting field of $g(x)$. Since $[E : F] = [E : F(\alpha)][F(\alpha) : F] = [E : F(\alpha)] \deg f$, we have $[E : F(\alpha)] \mid (n - 1)!$ by induction (since its degree is less than n since $\deg g = n - 1$), with $\deg f = n$, we have $[E : F] \mid n(n - 1)! = n!$, proving the result in this case.

Otherwise, suppose $f(x) \in F[x]$ is reducible. Write $f(x) = g(x)h(x)$ with $g(x), h(x) \in F[x]$, with $\deg g = m$, $\deg h = k$, with $1 < k, m < n$ with $m + k = n$. Let K be the splitting field of $g(x)$ over F . Since $\deg g = m$, by induction, $[K : F] \mid m!$. Since E is the splitting field of $h(x)$ over K , by induction

$[E : K] \mid k!$. Combining this, we get $[E : F] = [E : K][K : F] \mid m!k! \mid n!$ where the last divisibility follows since $\frac{n!}{m!k!} = \binom{n}{m} \in \mathbb{Z}$. ■

Chapter 5: More Field Theory

5.1 Prime Fields

Definition

The prime field of a field F is the intersection of all subfields of F .

Theorem 5.1

If F is a field, then its prime field is isomorphic to either $\mathbb{Q}$ or $\mathbb{Z}_p$.

Proof:

Let F_1 be a subfield of F and consider the unique ring homomorphism from $\mathbb{Z}$ to F_1 , $\chi : \mathbb{Z} \rightarrow F_1$ given by $\chi(n) = n1$ for all $n \in \mathbb{Z}$. Let $I = \ker \chi$. Since $\mathbb{Z}/I \cong \text{im } \chi$, a subring of F_1 , it is an integral domain, and so I is a prime ideal (we include the 0 ideal in that definition), so there are two cases:

- (1) If $I = \langle 0 \rangle$, then we have $\mathbb{Z} \subseteq F_1$ (identifying $\mathbb{Z}$ with its injective image under χ), so since F_1 is a field, $\mathbb{Q} = \text{Frac}(\mathbb{Z}) \subseteq F_1$ by minimality of the fraction field, and $\mathbb{Q}$ is a subfield of F .
- (2) Otherwise, $I = \langle p \rangle$, and so $\mathbb{Z}_p \cong \mathbb{Z}/I \cong \text{im } \chi \subseteq F_1$. ■

Definition

Given a field F , if its prime field is isomorphic to $\mathbb{Q}$ (respectively $\mathbb{Z}_p$) we say F has characteristic 0 (respectively characteristic p), denoted by $\text{ch}(F) = 0$ (respectively $\text{ch}(F) = p$).

Note that if $\text{ch}(F) = p$, for $a, b \in F$, $(a + b)^p = a^p + \binom{p}{1}a^{p-1}b + \cdots + \binom{p}{p-1}ab^{p-1} + b^p = a^p + b^p$ since $p \mid \binom{p}{i}$ since p is prime for $1 \leq i \leq p - 1$, and thus is 0. Using this,

Proposition 5.2

Let F be a field with $\text{ch}(F) = p$ and let $n \in \mathbb{N}$. Then the map $\phi : F \rightarrow F$ given by $u \mapsto u^{p^n}$ is an injective $\mathbb{Z}_p$ -homomorphism of fields. Moreover, if F is finite, then we automatically have surjectivity and so ϕ is a $\mathbb{Z}_p$ -isomorphism.

Proof:

Injectivity follows since a field homomorphism is injective or trivial, and we see that $\phi(1) = 1$ so it is injective. The rest is left as a Piazza exercise. ■

5.2 Formal Derivative and Repeated Roots

Definition

If F is a field, then the monomials $1, x, x^2, \dots$ form an F -basis of $F[x]$. Define a linear operator $D : F[x] \rightarrow F[x]$ by $D(1) = 0$ and $D(x^i) = ix^{i-1}$ for all $i \in \mathbb{N}$. Thus, for $f(x) = a_0 + a_1x + \cdots + a_nx^n \in F[x]$, we have $D(f(x)) = a_1 + 2a_2x + \cdots + na_nx^{n-1}$. We call $f'(x) := D(f(x))$ the formal derivative of f .

Remark

We have $D(f + g) = D(f) + D(g)$ by linearity, and $D(fg) = D(f)g + fD(g)$ (the Leibniz rule).

Theorem 5.3

Let F be a field and $f(x) \in F[x]$.

- (1) If $\text{ch}(F) = 0$, then $f'(x) = 0$ if and only if $f(x) = c$ for some $c \in F$.
- (2) If $\text{ch}(F) = p$, then $f'(x) = 0$ if and only if $f(x) = g(x^p)$ for some $g(x) \in F[x]$.

Proof:

For (1), the reverse direction is clear. In the forward direction, suppose $f(x) = a_0 + \cdots + a_n x^n$. Then $f'(x) = a_1 + 2a_2 x + \cdots + na_n x^{n-1}$, so if $f'(x) = 0$, we have $ia_i = 0$ for $i = 1, \dots, n$. Since $\text{ch}(F) = 0$, $i \neq 0$, we have $a_i = 0$ for $i = 1, \dots, n$, and thus $f(x) = a_0 \in F$.

For (2), in the reverse direction, write $g(x) = b_0 + b_1 x + \cdots + b_m x^m \in F[x]$. Then $f(x) := g(x^p) = b_0 + b_1 x^p + \cdots + b_m x^{pm}$, and thus $f'(x) = pb_1 x^{p-1} + \cdots + pmb_m x^{pm-1} = 0$ since $\text{ch}(F) = p$. In the forward direction, for $f(x) = a_0 + a_1 x + \cdots + a_n x^n$, $f'(x) = a_1 + 2a_2 x + \cdots + na_n x^{n-1} = 0$ implies that $ia_i = 0$ for $0 \leq i \leq n$. Since $\text{ch}(F) = p$, $ia_i = 0$ implies that $a_i = 0$ unless $p \mid i$. Thus $f(x) = a_0 + a_p x^p + a_{2p} x^{2p} + \cdots + a_{mp} x^{mp} = g(x^p)$ where $g(x) = a_0 + a_p x + a_{2p} x^2 + \cdots + a_{mp} x^m \in F[x]$. ■

Definition

Let E/F be a field extension and $f(x) \in F[x]$. We say $\alpha \in E$ is a repeated root of $f(x)$ if $(x - \alpha)^2 g(x)$ for some $g(x) \in E[x]$.

Theorem 5.4

Let E/F be a field extension, $f(x) \in F[x]$, and $\alpha \in E$. Then α is a repeated root of $f(x)$ if and only if $(x - \alpha) \mid f(x), f'(x)$, i.e., $(x - \alpha) \mid \gcd(f, f')$.

Proof:

In the forward direction, if $f(x) = (x - \alpha)^2 g(x)$, by the product rule, $f'(x) = 2(x - \alpha)g(x) + (x - \alpha)^2 g'(x) = (x - \alpha)(2g(x) + (x - \alpha)g'(x))$, so $(x - \alpha)$ divides both f and f' . In the reverse direction, suppose that $(x - \alpha)$ divides both f and f' . Write $f(x) = (x - \alpha)h(x)$ for $h(x) \in E[x]$. Then $f'(x) = h(x) + (x - \alpha)h'(x)$. Therefore, since α must be a root of $f'(x)$ since $x - \alpha \mid f'(x)$, we have $0 = f'(\alpha) = h(\alpha) + (\alpha - \alpha)h'(\alpha) = h(\alpha)$, so α is a root of $h(x)$, so $(x - \alpha)$ is one of its factors. It follows by taking out this factor that $f(x) = (x - \alpha)^2 g(x)$ for some $g(x) \in E[x]$. ■

Definition

Let F be a field and $f(x) \neq 0 \in F[x]$. We say $f(x)$ is separable over F if it has no repeated roots in any extension of F .

Example

$f(x) = (x - 2)(x + 9)$ is separable in $\mathbb{Q}[x]$.

Corollary 5.5

$f(x)$ is separable if and only if $\gcd(f, f') = 1$.

Proof:

Note that $\gcd(f, f') = 1$ if and only if $(x - \alpha) \mid \gcd(f, f')$ for α in some extension of F , since some degree 1 polynomial must divide both. The result follows by Theorem 5.4. ■

Remark

We notice that the condition of repeated roots depends on the extension of F , while the gcd condition above only depends on F .

Corollary 5.6

If $\text{ch}(F) = 0$, then every irreducible $r(x) \in F[x]$ is separable.

Proof:

Let $r(x) \in F[x]$ be irreducible. Since it is irreducible, its only monic divisors are 1 and $r(x)$, so $\gcd(r, r') = r$ if $r' \mid r$, which occurs only if $r' = 0$ by irreducibility, and otherwise $\gcd(r, r') = 1$ if $r' \neq 0$. Suppose that $r(x)$ is not separable. Then by Corollary 5.5, $\gcd(r, r') \neq 1$. Thus $r' = 0$, so r is constant by Theorem 5.3, contradicting irreducibility. ■

Example

The p th cyclotomic polynomial $\Phi_p(x) = x^{p-1} + \cdots + x + 1$ is irreducible over $\mathbb{Q}$, and hence separable. We note that the roots of $\Phi_p(x)$ are $\zeta_p, \zeta_p^2, \dots, \zeta_p^{p-1}$, which are all distinct.

5.3 Finite Fields

Given a field F , let $F^* = F \setminus \{0\}$ be the multiplicative group of nonzero elements of F .

Proposition 5.7

If F is a finite field, then $\text{ch}(F) = p$ for some prime p and $|F| = p^n$ for some $n \in \mathbb{N}$.

Proof:

Because F is finite, it cannot contain (a field isomorphic to) $\mathbb{Q}$ as a subfield, and thus by Theorem 5.1 its prime field must be $\mathbb{Z}_p$ for some prime p . But then since F is finite, as a vector space over $\mathbb{Z}_p$ it must be finite dimensional, and so $F \cong \mathbb{Z}_p^n$ as a $\mathbb{Z}_p$ -vector space, and thus $|F| = p^n$. ■

Theorem 5.8

Let F be a field, and G a finite subgroup of F^* . Then G is a cyclic group. In particular, if F is a finite field, then F^* is a cyclic group.

Proof:

Without loss of generality, assume $G \neq \{1\}$. Because G is a finite abelian group, by the fundamental theorem of finitely generated abelian groups, we have $G \cong \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} \times \cdots \times \mathbb{Z}/n_r\mathbb{Z}$ with $n_1 \mid n_2 \mid \cdots \mid n_r$. Since $n_r(\mathbb{Z}/n_1\mathbb{Z} \times \cdots \times \mathbb{Z}/n_r\mathbb{Z}) = 0$, it follows that every $u \in G$ is a root of $x^{n_r} - 1 \in F[x]$. Since the polynomial has at most n_r distinct roots in F , we have $r = 1$ and $G \cong \mathbb{Z}/n_r\mathbb{Z}$. ■

Corollary 5.9

If F is finite field, then F is a simple extension of $\mathbb{Z}_p$, that is, $F = \mathbb{Z}_p(u)$ for some $u \in F$. ■

Theorem 5.10

Let p be a prime and $n \in \mathbb{N}$.

(1) F is a finite field with $|F| = p^n$ if and only if F is splitting field of $x^{p^n} - x \in \mathbb{Z}_p[x]$.

(2) Let F be a finite field with $|F| = p^n$. Let $m \in \mathbb{N}$ with $m \mid n$. Then F contains a unique subfield K with $|K| = p^m$.

Proof:

Proving (1) in the forward direction, suppose $|F| = p^n$. Then $|F^*| = p^n - 1$. Then every $u \in F^*$ satisfies $u^{p^n-1} = 1$, and thus it is a root of $x(x^{p^n-1} - 1) = x^{p^n} - x \in \mathbb{Z}_p[x]$. Since $0 \in F$ is also a root of $x^{p^n} - x$, the polynomial $x^{p^n} - x$ has p^n distinct roots in F , so it splits over F . Because F only contains these roots and nothing else, it must be a minimal field that splits over F , and so F is a splitting field of $x^{p^n} - x$ over $\mathbb{Z}_p$. Conversely, suppose that F is a splitting field of $f(x) := x^{p^n} - x \in \mathbb{Z}_p[x]$. Since $\text{ch}(F) = p$, we have $f'(x) = -1$. Therefore $\gcd(f, f') = 1$, so by Corollary 5.5, $f(x)$ has p^n distinct roots in F . Let E be the set of all roots of $f(x)$ in F . Let $\phi : F \rightarrow F$ be given by $u \mapsto u^{p^n}$. For $u \in F$, u is a root of $f(x)$, and thus in E , if and only if $\phi(u) = u$. But the set satisfying the condition $\phi(u) = u$ is closed under field operations (since ϕ is a homomorphism), so E is a subfield of F of order p^n , containing $\mathbb{Z}_p$ (since all $u \in \mathbb{Z}_p$ satisfy $u^{p^n} = u$ by Fermat's Little Theorem). Since F is a splitting field of $f(x)$, it is generated over $\mathbb{Z}_p$ by the roots of $f(x)$, that is, $F = \mathbb{Z}_p(E) = E$, proving the desired result since $|E| = p^n$.

Proving (2), we recall from high school $x^{ab} - 1 = (x^a - 1)(x^{ab-a} + \cdots + x^a + 1)$. Thus if $n = mk$, we have $p^n - 1 = p^{mk} - 1 = (p^m - 1) \cdots$ by the above identity, so we have $x^{p^n} - x = x(x^{p^n-1} - 1) = x(x^{p^m-1} - 1)g(x)$ for some $g(x) \in \mathbb{Z}_p[x]$. Since $(x^{p^n} - x)$ splits over F by (1), so does $x^{p^m} - x$ because it is a factor. Let K be the set of roots of $x^{p^m} - x$. We have $|K| = p^m$ by the same argument as in (1), and similarly, K is a field. Moreover, note that if $\tilde{K} \subseteq F$ is a subfield with $|\tilde{K}| = p^m$, then by (1), $\tilde{K}$ must be a subset of the roots of $x^{p^m} - x$, and thus $\tilde{K} \subseteq K$, and we have equality because the sets are the same (finite) size. It follows that K is the unique subfield of F of order p^m . ■

As a direct consequence of Theorem 5.10 and Corollary 4.5, we have:

Corollary 5.11 (E.H. Moore)

Let p be a prime and $n \in \mathbb{N}$. Then any two finite fields of order p^n are isomorphic. We denote such a field by $\mathbb{F}_{p^n}$. ■

Corollary 5.12

Let F be a finite field with $\text{ch}(F) = p$. Then,

- (1) $F = F^p = \{x^p : x \in F\}$
- (2) Every irreducible polynomial $r(x) \in F[x]$ is separable.

Proof:

Proving (1), every finite field $F = \mathbb{F}_{p^n}$ is the splitting field of $x^{p^n} - x$ over $\mathbb{F}_p$ for some prime p and $n \in \mathbb{N}$. Then for every $a \in F$,

$$a = a^{p^n} = (a^{p^{n-1}})^p$$

Since $a^{p^{n-1}} \in F$, $F = F^p$.

Proving (2), let $r(x) \in F[x]$ be irreducible. Then $\gcd(r, r') = 1$ if $r' \neq 0$, and is equal to r if $r' = 0$. Suppose $r(x)$ is not separable. Then by Corollary 5.5, $\gcd(r, r') \neq 1$, so $r'(x) = 0$. By Theorem 5.3, this implies that $r(x) = a_0 + a_1 x^p + \cdots + a_m x^{mp} \in F[x]$. Since $F = F^p$ by (1), we can write $a_i = b_i^p$ with $b_i \in F$ for each i . Thus $r(x) = b_0^p + b_1^p x^p + \cdots + b_m^p x^{mp} = (b_0 + b_1 x + \cdots + b_m x^m)^p$ (by characteristic p), which is not irreducible because we factored it nontrivially, a contradiction, and thus $r(x)$ must be separable. ■

We've now seen in characteristic 0 and characteristic p if our field is finite, irreducibility implies separability. What about in the infinite $\text{ch}(F) = p$ case?

Example

Let $\text{ch}(F) = p$. Consider $f(x) = x^p - a$. Since $f'(x) = px^{p-1} = 0$, we have $\gcd(f, f') \neq 1$ so this is not separable by Corollary 5.5. Consider F^p , which is a subfield of F always in characteristic p .

(1) If $a \in F^p$, say $a = b^p$ for some $b \in F$, then $f(x) = x^p - b^p = (x - b)^p \in F[x]$, which is not separable, and also reducible in $F[x]$.

(2) Suppose $a \notin F^p$. Let E/F be an extension where $f(x)$ has a root $\beta \in E$. Hence we have $\beta^p - a = 0$, so $a = \beta^p \notin F^p$, so we have $\beta \notin F$. But then $f(x) = (x - \beta)^p$ which is not separable. We claim this implies $f(x)$ is irreducible in $F[x]$. Suppose $x^p - a = g(x)h(x)$, where $g(x), h(x) \in F[x]$ are monic. We have seen $x^p - a = (x - \beta)^p$, and so we must have $g(x) = (x - \beta)^r$ and $h(x) = (x - \beta)^s$ for $r, s \in \mathbb{N} \cup \{0\}$ such that $r + s = p$. Write $g(x) = x^r - r\beta x^{r-1} + \cdots$, and so $r\beta \in F$, and because $\beta \notin F$, this implies $r\beta$ vanishes so $r = 0$ in F , so we see either $r = p, s = 0$ or $r = 0, s = p$ as an integer. Thus $f(x)$ is irreducible in $F[x]$, since either $g(x) = f(x)$ or $h(x) = f(x)$ (and the other is 1).

Chapter 6: Solvable Groups and Automorphism Groups

6.1 Solvable Groups

Definition

A group G is solvable if there exists a tower of subgroups $G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_m = \{1\}$ with $G_{i+1} \trianglelefteq G_i$ and G_i/G_{i+1} abelian for all $0 \leq i \leq m-1$.

Remark

G_{i+1} is not necessarily a normal subgroup of G , only a normal subgroup of G_i . However if G_{i+1} is normal in G , we get $G_{i+1} \trianglelefteq G_i$ for free.

Example

Consider the symmetric group S_4 , with A_4 the alternating group and $V \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ the Klein 4-group, which is a subgroup of A_4 , and normal in S_4 . We thus have the tower of subgroups $S_4 \supseteq A_4 \supseteq V \supseteq \{1\}$,

where $S_4/A_4 \cong \mathbb{Z}_2$, $A_4/V \cong \mathbb{Z}_3$, which are all abelian. Thus S_4 is solvable.

Recall the following theorems:

Theorem 6.1 (Second Isomorphism Theorem)

Let H and K be subgroups of a group G with $K \trianglelefteq G$. Then HK is a subgroup of G , with $K \trianglelefteq HK$, $H \cap K \trianglelefteq H$, and $HK/K \cong H/(H \cap K)$. ■

Theorem 6.2 (Third Isomorphism Theorem)

Let $K \subseteq H \subseteq G$ be groups with $K \trianglelefteq G$ and $H \trianglelefteq G$. Then $H/K \trianglelefteq G/K$ and $(G/K)/(H/K) \cong G/H$. ■

Theorem 6.3

Let G be a solvable group.

- (1) If H is a subgroup of G , then H is solvable.
- (2) Let N be a normal subgroup of G . Then G/N is solvable.

Proof:

Since G is a solvable group, there exists a tower of subgroups $G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_m = \{1\}$ with $G_{i+1} \trianglelefteq G_i$ and G_i/G_{i+1} abelian for all $0 \leq i \leq m-1$.

(1) For each i , define $H_i = H \cap G_i$. This gives a tower $H = H_0 \supseteq H_1 \supseteq \cdots \supseteq H_m = \{1\}$, where the tower satisfies $H_{i+1} \trianglelefteq H_i$. Note that both H_i and G_{i+1} are subgroups of G_i and $H_{i+1} = H \cap G_{i+1} = H_i \cap G_{i+1}$, so by the Second Isomorphism Theorem (applied to G_i), we have

$$H_i/H_{i+1} = H_i/(H_i \cap G_{i+1}) \cong H_i G_{i+1}/G_{i+1} \subseteq G_i/G_{i+1}$$

Since G_i/G_{i+1} is abelian, this shows H_i/H_{i+1} must be abelian, and thus H is solvable.

(2) Consider the towers $G = G_0N \supseteq G_1N \supseteq \cdots \supseteq G_mN = N$, and $G/N = G_0N/N \supseteq G_1N/N \supseteq \cdots \supseteq G_mN/N = \{1\}$. Since $G_{i+1} \trianglelefteq G_i$, and $N \trianglelefteq G$, we have that $G_{i+1}N \trianglelefteq G_iN$, which implies that $G_{i+1}N/N \trianglelefteq G_iN/N$. By the Third Isomorphism Theorem, we have $(G_iN/N)/(G_{i+1}N/N) \cong G_iN/G_{i+1}N$, and by the Second Isomorphism Theorem ($H = G_i, K = G_{i+1}N$), $G_iN/G_{i+1}N \cong G_i/(G_i \cap G_{i+1}N)$. Consider the quotient homomorphism $G_i \rightarrow G_i/(G_i \cap G_{i+1}N)$, which is surjective. Since $G_{i+1} \subseteq G_i \cap G_{i+1}N$, it induces a surjective homomorphism $G_i/G_{i+1} \rightarrow G_i/(G_i \cap G_{i+1}N)$. Since G_i/G_{i+1} is abelian, by surjectivity of this homomorphism, this shows that $G_i/(G_i \cap G_{i+1}N) \cong (G_iN/N)/(G_{i+1}N/N)$ is abelian, and so G/N is solvable. ■

Theorem 6.4

Let N be a normal subgroup of G . If both N and G/N are solvable, then G is solvable. In particular, a direct product of finitely many solvable groups is solvable.

Proof: Since N is solvable, we have a tower of N_i 's, and denoting with a bar any quotient by N , we have an abelian tower $G/N = \overline{G} = \overline{G}_0 \supseteq \cdots \supseteq \overline{G}_m = N/N = \{1\}$. Let $\text{Sub}_N(G)$ denote the subgroups of G containing N . Consider the map $\sigma : \text{Sub}_N(G) \rightarrow \text{Sub}(G/N)$, $H \mapsto H/N$. For all i , define $G_i = \sigma^{-1}(\overline{G}_i)$. Since $N \trianglelefteq G$ and $\overline{G}_{i+1} \trianglelefteq \overline{G}_i$, we have $G_{i+1} \trianglelefteq G_i$, and by the third isomorphism theorem, we have $G_i/G_{i+1} \cong \overline{G}_i/\overline{G}_{i+1}$. It follows that $G = G_0 \supseteq \cdots \supseteq G_r = N = N_0 \supseteq N_1 \supseteq \cdots \supseteq N_m = \{1\}$ is an abelian tower, so G is solvable. ■

Example

S_4 contains subgroups isomorphic to S_3 and S_2 . Since S_4 is solvable, by Theorem 6.3, S_3 and S_2 are solvable.

Definition

A group is simple if it is not trivial and has no normal subgroups except $\{1\}$ and G .

Example

One can show A_5 is simple (bonus 4, pmath 347). Since $A_5 \supseteq \{1\}$ is the only tower and $A_5/\{1\}$ is not

abelian, A_5 is not solvable. Thus by Theorem 6.3, S_5 is also not solvable. Moreover, since all S_n with $n \geq 5$ contain a subgroup isomorphic to S_5 , they are not solvable either by Theorem 6.3.

Corollary 6.5

Let G be a finite solvable group. Then there exists an abelian tower $G = G_0 \supseteq \cdots \supseteq G_m = \{1\}$, where each G_i/G_{i+1} is cyclic.

Proof:

If G is solvable, we have an abelian tower $G = G_0 \supseteq \cdots \supseteq G_n = \{1\}$. Consider $A = G_i/G_{i+1}$, which is a finite abelian group. We have $A \cong C_{k_1} \times C_{k_2} \times \cdots \times C_{k_r}$ where C_k is a cyclic group of order k . Picking any of these, say C_{k_1} , gives a cyclic subgroup of A and thus corresponds to a normal subgroup of G_i containing G_{i+1} , call it $H_{i,1}$, where $H_{i,1}/G_{i+1}$ is cyclic. Repeat this process to get a cyclic tower $G_{i+1} = H_{i,n} \supseteq \cdots \supseteq H_{i,1} \supseteq H_{i,0} = G_i$ (each quotient is cyclic), where it terminates since this always decreases the index of the subgroup, and then do this for every i . ■

Remark

By the Chinese remainder theorem, we can further require the quotient G_i/G_{i+1} to be a cyclic group of prime order.

6.2 Automorphism Groups

Definition

Let E/F be a field extension. If $\psi : E \rightarrow E$ is an automorphism of E that satisfies $\psi|_F = 1_F$, we say ψ is an F -automorphism of E . Using composition of the group operation, the set of F -automorphisms of E , $\text{Aut}_F(E)$ is a group, called the automorphism group of E/F .

Lemma 6.6

Let E/F be a field extension, with $f(x) \in F[x]$ and $\psi \in \text{Aut}_F(E)$. If $\alpha \in E$ is a root of $f(x)$, then $\psi(\alpha)$ is also a root of $f(x)$.

Proof:

Write $f(x) = a_0 + a_1x + \cdots + a_nx^n \in F[x]$. We have

$$\begin{aligned} f(\psi(\alpha)) &= a_0 + a_1\psi(\alpha) + \cdots + a_n(\psi(\alpha))^n = \psi(a_0 + \psi(a_1)\psi(\alpha) + \cdots + \psi(a_n)(\psi(\alpha))^n) \\ &= \psi(a_0 + a_1\alpha + \cdots + a_n\alpha^n) = \psi(f(\alpha)) = \psi(0) = 0 \quad \blacksquare \end{aligned}$$

Lemma 6.7

Let $E = F(\alpha_1, \dots, \alpha_n)$. For $\psi_1, \psi_2 \in \text{Aut}_F(E)$, if $\psi_1(\alpha_i) = \psi_2(\alpha_i)$ for all $i = 1, \dots, n$, then $\psi_1 = \psi_2$.

Proof:

Note that for $\alpha \in E$, α is of the form $\frac{f(\alpha_1, \dots, \alpha_n)}{g(\alpha_1, \dots, \alpha_n)}$ where $f(x_1, \dots, x_n), g(x_1, \dots, x_n) \in F[x_1, \dots, x_n]$ with $g \neq 0$. The lemma then follows immediately because we see an F -automorphism of E is determined by its action on the α_i 's. ■

Corollary 6.8

If E/F is a finite extension, then $\text{Aut}_F(E)$ is a finite group.

Proof:

Any finite extension can be written as $E = F(\alpha_1, \dots, \alpha_n)$ by Theorem 3.5, and by Lemma 6.7, the action of an F -automorphism of E is determined by its action on the α_i 's. But by Lemma 6.6, for all i , for any $\psi \in \text{Aut}_F(E)$, $\psi(\alpha_i)$ is a root of the minimal polynomial of α_i over F , thus there are only finitely many choices for ψ 's action on each α_i , and so only finitely many ψ 's. ■

Remark

The converse of the above is false, for example $\mathbb{R}/\mathbb{Q}$ is an infinite extension, but $\text{Aut}_{\mathbb{Q}}(\mathbb{R}) = \{1\}$.

6.3 Automorphism Groups of Splitting Fields

Definition

Let F be a field and $f(x) \in F[x]$. The automorphism group of $f(x)$ over F is the group $\text{Aut}_F(E)$ where E is the splitting field of $f(x)$ over F .

By Theorem 4.4 and A4Q2, we have

Theorem 6.9

Let E/F be a splitting field of a nonzero polynomial $f(x) \in F[x]$. We have $|\text{Aut}_F(E)| \leq [E : F]$ and equality holds if and only if every irreducible factor of $f(x)$ is separable. ■

Theorem 6.10

If $f(x) \in F[x]$ has n distinct roots in the splitting field E , then $\text{Aut}_F(E)$ is isomorphic to a subgroup of S_n . In particular $|\text{Aut}_F(E)|$ divides $n!$.

Proof:

Let $X = \{\alpha_1, \dots, \alpha_n\}$ be the distinct roots of $f(x)$ in E . By Lemma 6.6, if $\psi \in \text{Aut}_F(E)$, then $\psi(X) = X$. Let $\psi|_X$ be the restriction of ψ in X , and $S_X \cong S_n$ the permutation group of X . The map $\text{Aut}_F(E) \rightarrow S_X$ $\psi \mapsto \psi|_X$ is a group homomorphism. Moreover, by Lemma 6.7, it is injective. Therefore, $\text{Aut}_F(E)$ is isomorphic to a subgroup of S_n . ■

Example

Let $f(x) = x^3 - 2 \in \mathbb{Q}[x]$, and $E/\mathbb{Q}$ the splitting field of $f(x)$. Then $E = \mathbb{Q}(\sqrt[3]{2}, \zeta_3)$, with $[E : \mathbb{Q}] = [E : \mathbb{Q}(\sqrt[3]{2})][\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 2 \cdot 3 = 6$. Since $\text{ch}(\mathbb{Q}) = 0$, $f(x)$ is separable (it is irreducible). By Theorem 6.9, $|\text{Aut}_{\mathbb{Q}}(E)| = [E : \mathbb{Q}] = 6$. Since $f(x)$ has 3 distinct roots in E , by Theorem 6.10, $\text{Aut}_{\mathbb{Q}}(E)$ is a subgroup of S_3 , but the only subgroup of S_3 of order 6 is S_3 itself, so $\text{Aut}_{\mathbb{Q}}(E) \cong S_3$.

Example

Let F be a field with $\text{ch}(F) = p$, and $F^p \neq F$. Let $f(x) = x^p - a$ with $a \in F \setminus F^p$. Let E/F be a splitting field of $f(x)$. We have seen that $f(x)$ is irreducible in $F[x]$, and $f(x) = (x - \beta)^p$ for some $\beta \in E \setminus F$. Thus $E = F(\beta)$. Since β can only map to β , $\text{Aut}_F(E)$ is trivial. On the other hand $[E : F] = \deg(f) = p$, and so $|\text{Aut}_F(E)| \neq [E : F]$.

Definition

Let E/F be a field extension and let $\psi \in \text{Aut}_F(E)$. Define $E^\psi = \{a \in E : \psi(a) = a\}$ which is a subfield of E containing F . We call E^ψ the fixed field of ψ . If $G \subseteq \text{Aut}_F(E)$, the fixed field of G is defined by $E^G = \bigcap_{\psi \in G} E^\psi = \{a \in E : \psi(a) = a \ \forall \ \psi \in G\}$.

Theorem 6.11

Let $f(x) \in F[x]$ be a polynomial in which every irreducible factor is separable. Let E/F be a splitting field of $f(x)$. If $G = \text{Aut}_F(E)$, then $E^G = F$.

Proof:

Set $L = E^G$. Since $F \subseteq L$, we have $\text{Aut}_L(E) \subseteq \text{Aut}_F(E)$. On the other hand, if $\psi \in \text{Aut}_F(E)$, by definition of L , for all $a \in L$, we have $\psi(a) = a$. This implies $\psi \in \text{Aut}_L(E)$. This implies $\psi \in \text{Aut}_L(E)$, and thus $\text{Aut}_F(E) = \text{Aut}_L(E)$. Note that since $f(x)$ is separable over F and splits over E , $f(x)$ is also separable over L and has E as its splitting field over L . Thus by Theorem 6.9, we have

$$|\text{Aut}_F(E)| = [E : F], |\text{Aut}_L(E)| = [E : L]$$

It follows that $[E : F] = [E : L]$. Since $[E : F] = [E : L][L : F]$, this implies $[L : F] = 1$, so $L = F$. ■

Chapter 7: Separable Extensions and Normal Extensions

7.1 Separable Extensions

Definition

Let E/F be an algebraic field extension. For $\alpha \in E$, let $p(x) \in F[x]$ be the minimal polynomial of α . We say α is separable over F if $p(x)$ is separable. If every $\alpha \in E$ is separable, we say E/F is separable.

Example

In $\text{ch}(F) = 0$, every irreducible polynomial $p(x)$ is separable, so every algebraic extension E/F is separable.

Theorem 7.1

Let E/F be the splitting field of $f(x) \in F[x]$. If every irreducible factor of $f(x)$ is separable, then E/F is separable.

Proof:

Let $\alpha \in E$ and let $p(x) \in F[x]$ be the minimal polynomial of α . Let $\{\alpha_1, \dots, \alpha_n\}$ be the set of all distinct roots of $p(x)$ in E . Let $\tilde{p}(x) = (x - \alpha_1) \dots (x - \alpha_n)$. We claim $\tilde{p}(x) \in F[x]$. Let $G = \text{Aut}_F(E)$, and let $\psi \in G$. Since ψ is an automorphism, $\psi(\alpha_i) \neq \psi(\alpha_j)$ if $i \neq j$. By Lemma 6.6, ψ permutes $\alpha_1, \dots, \alpha_n$. Thus by extending $\psi : E \rightarrow E$ uniquely to $\psi : E[x] \rightarrow E[x]$ with $x \mapsto x$, we have $\psi(\tilde{p}(x)) = (x - \psi(\alpha_1)) \dots (x - \psi(\alpha_n))$, but since ψ permutes the α_i 's, this is just $(x - \alpha_1) \dots (x - \alpha_n)$ possibly rearranged, so $\psi(\tilde{p}(x)) = \tilde{p}(x)$, from which it follows that $\tilde{p}(x) \in E^\psi[x]$. Since $\psi \in G$ was arbitrary, it shows $\tilde{p}(x) \in E^G[x]$. Since E/F is the splitting field of $f(x)$ whose irreducible factors are separable, by Theorem 6.11, $E^G = F$ so $\tilde{p}(x) \in F[x]$. By minimality, we have $p(x) \mid \tilde{p}(x)$, and because we wrote down the roots of $\tilde{p}(x)$ as a subset of the roots of $p(x)$, we have $\tilde{p}(x) \mid p(x)$. Because they are both monic, this shows $p(x) = \tilde{p}(x)$. But $\tilde{p}(x)$ is separable (because we only took distinct roots in its definition), so $p(x)$ is separable. ■

Corollary 7.2

Let E/F be a finite extension and suppose $E = F(\alpha_1, \dots, \alpha_n)$. If each α_i is separable over F , then E/F is separable.

Proof:

Let $p_i(x) \in F[x]$ be the minimal polynomial of some α_i for $i = 1, \dots, n$. Let $f(x) = p_1(x) \dots p_n(x)$ each $p_i(x)$ separable by hypothesis. Let L be the splitting field of $f(x)$ over F . By Theorem 7.1, L/F is separable, but $E = F(\alpha_1, \dots, \alpha_n)$ is a subfield of L , so E is also separable over F . ■

Corollary 7.3

Let E/F be an algebraic extension, and let L be the set of all $\alpha \in E$ which are separable over F . Then L is a field.

Proof:

Let $\alpha, \beta \in L$. Then $\alpha \pm \beta, \alpha\beta, \alpha/\beta$ ($\beta \neq 0$) are all in $F(\alpha, \beta)$. By Corollary 7.2, $F(\alpha, \beta)$ is separable and hence is contained in L , from which it follows that $\alpha \pm \beta, \alpha\beta, \alpha/\beta$ ($\beta \neq 0$) $\in L$. ■

We've seen that a finite extension is a composition of simple extensions, we'll generalize this:

Definition

Let $E = F(\gamma)$ be a simple extension. We call γ a primitive element of E/F .

Theorem 7.4 (Primitive Element Theorem)

If E/F is a finite separable extension, then $E = F(\gamma)$ for some $\gamma \in E$. In particular, if $\text{ch}(F) = 0$, then any finite extension E/F is a simple extension.

Proof:

We have seen in Corollary 5.9 that a finite extension of a finite field is always simple. Thus without loss

of generality, assume F is an infinite field. Since $E = F(\alpha_1, \dots, \alpha_n)$, it suffices by induction to consider the case $E = F(\alpha, \beta)$, with $\alpha, \beta \notin F$. We claim there exists $\lambda \in F$ such that $\beta \in F(\alpha + \lambda\beta)$, with the idea to take $\gamma = \alpha + \lambda\beta$. Once we know the claim holds, then clearly $F(\gamma) = F(\alpha + \lambda\beta) \subseteq F(\alpha, \beta)$, and we will have shown $\beta \in F(\gamma)$, and then moreover $\gamma - \lambda\beta = \alpha$, so $\alpha \in F(\gamma)$ since $\gamma, \lambda, \beta \in F(\gamma)$ (since $\lambda \in F$), showing $F(\alpha, \beta) \subseteq F(\gamma)$, and we have the result. To prove the claim, let $a(x)$ and $b(x)$ be the minimal polynomials of α and β over F respectively. Since $\beta \notin F$, $\deg(b) > 1$. Thus there exists a root $\tilde{\beta}$ of $b(x)$ such that $\tilde{\beta} \neq \beta$. Choose $\lambda \in F$ such that $\lambda \neq \frac{\tilde{\alpha} - \alpha}{\beta - \tilde{\beta}}$ for all roots $\tilde{\alpha}$ of $a(x)$ and all roots $\tilde{\beta}$ of $b(x)$ with $\tilde{\beta} \neq \beta$ in some splitting field of $a(x)b(x)$ over F , which is only possible because F is infinite but $a(x)b(x)$ only has finitely many roots. Let $\gamma = \alpha + \lambda\beta$. Consider $h(x) = a(\gamma - \lambda x) \in F(\gamma)[x]$. Then $h(\beta) = a(\gamma - \lambda\beta) = a(\alpha) = 0$, so β is a root of $h(x)$. However, for any $\tilde{\beta} \neq \beta$, we have $\gamma - \lambda\tilde{\beta} = \alpha + \lambda(\beta - \tilde{\beta}) \neq \tilde{\alpha}$ by the choice of λ , but the $\tilde{\alpha}$'s are all the roots of $a(x)$, so this shows $\tilde{\beta}$ is not a root of $h(x)$. Thus, $h(x)$ and $b(x)$ have β as a common root, but share no other roots in any extension of $F(\gamma)$. Let $b_1(x)$ be the minimal polynomial of β over $F(\gamma)$. Then $b_1(x)$ divides both $h(x)$ and $b(x)$. Since E/F is separable with $b(x) \in F[x]$ irreducible, $b(x)$ has distinct roots, so therefore $b_1(x)$ also must have distinct roots. But $b_1(x)$'s roots are common to $h(x)$ and $b(x)$, so since those two only share β as a root in common, and it is not repeated, we have $b_1(x) = x - \beta$. Since $b_1(x) \in F(\gamma)[x]$, we get $\beta \in F(\gamma)$, which is what we wanted to show. ■

7.2 Normal Extensions

Definition

Let E/F be an algebraic extension. We say E/F is a normal extension, if for any irreducible polynomial $p(x) \in F[x]$, either $p(x)$ has no roots in E or $p(x)$ has all roots in E . In other words, if $p(x)$ has a root in E , $p(x)$ splits over E .

Example

Let $\alpha \in \mathbb{R}$ with $\alpha^4 = 5$. Since the roots of $x^4 - 5$ are $\pm\alpha, \pm\alpha i$, and $Q(\alpha)$ is real, $Q(\alpha)/\mathbb{Q}$ is not normal. What about if we try adding i to our extension to solve this problem? Let $\beta = (1 + i)\alpha$. We claim $Q(\beta)/\mathbb{Q}$ is also not normal. Notice that $\beta^2 = 2i\alpha^2, \beta^4 = -4\alpha^4 = -20$, and $\pm\beta$ and $\pm i\beta$ all satisfy $x^4 = -20$, to show $Q(\beta)$ is not normal, it suffices to show $i \notin Q(\beta)$. Since our calculations above show the minimal polynomial of β over $\mathbb{Q}$ is $p(x) = x^4 + 20$, we have $[Q(\beta) : \mathbb{Q}] = 4$, and the roots of $p(x)$ are $\pm\beta, \pm i\beta$. Since the minimal polynomial of α is $x^4 - 5$, we have $[Q(\alpha) : \mathbb{Q}] = 4 = [Q(\beta) : \mathbb{Q}]$, so if $\alpha \in Q(\beta)$, we have $Q(\alpha) = Q(\beta)$, which is not possible since $Q(\alpha)$ is real, but $Q(\beta)$ is not. But then, if $i \in Q(\beta)$, then $\alpha = \frac{\beta}{1+i}$ is in $Q(\beta)$, which is impossible. Therefore, our argument shows $Q(\beta)/\mathbb{Q}$ cannot be normal.

Theorem 7.5

A finite extension E/F is normal if and only if it is the splitting field of some $f(x) \in F[x]$.

Proof:

In the forward direction, suppose that E/F is normal, and write $E = F(\alpha_1, \dots, \alpha_n)$. Let $p_i(x) \in F[x]$ be the minimal polynomial of α_i for each i . Define $f(x) = p_1(x) \dots p_n(x)$. Since E/F is normal, each $p_i(x)$ splits over E (since it contains at least one root of each), so $f(x)$ splits over E . For each i , let $\alpha_i = \alpha - i, 1, \alpha_{i,2}, \dots, \alpha_{i,r_i}$ be the roots of $p_i(x)$ in E . Then $E = F(\alpha_1, \dots, \alpha_n) = F(\alpha_{1,1}, \dots, \alpha_{1,r_1}, \dots, \alpha_{2,1}, \dots, \alpha_{2,r_2}, \dots, \alpha_{n,1}, \dots, \alpha_{n,r_n})$, and this is the splitting field of $f(x)$ over E . Conversely, let E/F be the splitting field of $f(x) \in F[x]$. Let $p(x) \in F[x]$ be irreducible and suppose it has a root $\alpha \in E$. Let K/E be the splitting field of $p(x)$ over E . Write $p(x) = c(x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_n)$, $c \neq 0 \in F$, where $K = E(\alpha_1, \dots, \alpha_n)$. Since $F(\alpha) \cong F[x]/\langle p(x) \rangle \cong F(\alpha_2)$, we have the F -isomorphism $\theta : F(\alpha_1) \rightarrow F(\alpha_2)$ given by fixing F and mapping α to α_2 . Note that $p(x)f(x) \in F[x] \subseteq F(\alpha)[x]$ and $p(x)f(x) \in F(\alpha_2)[x]$. Thus we view K as the splitting field of $p(x)f(x)$ over $F(\alpha)$ and $F(\alpha_2)$. Thus by Theorem 4.4, there exists an isomorphism $\psi : K \rightarrow K$ extending θ . In particular, $\psi \in \text{Aut}_F(K)$. Thus, ψ permutes the roots of $f(x)$ by Lemma 6.6, and thus fixes E because E is just F with the roots of $f(x)$.

adjoined. Thus, for $\alpha \in E$, $\alpha_2 = \psi(\alpha) \in E$. A similar argument shows $\alpha_i \in E$ for all $3 \leq i \leq n$. Since all α_i 's are in E , this shows $K = E(\alpha_1, \dots, \alpha_n) = E$, and so $p(x)$ splits over E . It follows that E/F is normal. ■

Example

Every quadratic extension is normal: Let E/F be a field extension with $[E : F] = 2$. For $\alpha \in E \setminus F$, we have $E = F(\alpha)$. Let $p(x) = x^2 + ax + b$ the minimal polynomial of α over F . If β is another root of $p(x)$, then $p(x) = (x - \alpha)(x - \beta) = x^2 - (\alpha + \beta)x + \alpha\beta$, so $\beta = -a - \alpha$. Since $a \in F$, this shows $\beta \in F$. Hence E/F is the splitting field of $p(x)$, so it is normal.

Example

Consider the extension $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}$. This is not normal, since the irreducible polynomial $p(x) = x^4 - 2$ has $\sqrt[4]{2}$ as a root, but does not split over $\mathbb{Q}(\sqrt[4]{2})$ since it has nonreal roots. Note that $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}$ is made up of two quadratic extensions, $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$, which are both normal. Thus, we see if E/K and K/F are normal extensions, E/F is not always normal.

Proposition 7.6

If E/F is a normal extension, and K is an intermediate field $F \subseteq K \subseteq E$, then E/K is normal.

Proof:

Let $p(x) \in K[x]$ be irreducible and has a root $\alpha \in E$, so we may assume by dividing out the leading coefficient if necessary that $p(x)$ is the minimal polynomial of α . Let $f(x) \in F[x] \subseteq K[x]$ be the minimal polynomial of α over F . Because $F[x] \subseteq K[x]$, by minimality of $p(x)$, we have $p(x) \mid f(x)$. Since E/F is normal, we thus have that $f(x)$ splits over E , so therefore so too does its factor $p(x)$, which shows E/K is normal. ■

Example

Even under the hypotheses of the above proposition, K/F is not always normal. For example, take $F = \mathbb{Q}$, $K = \mathbb{Q}(\sqrt[4]{2})$, $E = \mathbb{Q}(\sqrt[4]{2}, i)$. Then E/F is the splitting field of $x^4 - 2$, hence normal, and E/K is normal, but $K/\mathbb{Q}$ is not normal as seen in the above example.

Proposition 7.7

Let E/F be a finite normal extension, with $\alpha, \beta \in E$. The following are equivalent:

- (1) There exists $\psi \in \text{Aut}_F(E)$ such that $\psi(\alpha) = \beta$.
- (2) The minimal polynomial of α and β over F are the same.

In this case, we say α and β are conjugate over F .

Proof:

Proving (1) implies (2), let $p(x)$ be the minimal polynomial of α over F , and $\psi \in \text{Aut}_F(E)$ with $\psi(\alpha) = \beta$. By Lemma 6.6, β is also a root of $p(x)$. Since $p(x)$ is monic and irreducible, it must too be β 's minimal polynomial over F , so α and β have the same minimal polynomial.

Proving (2) implies (1), suppose that α and β share a minimal polynomial $p(x) \in F[x]$. Since $F(\alpha) \cong F[x]/\langle p(x) \rangle \cong F(\beta)$, we have an F -isomorphism $\theta : F(\alpha) \rightarrow F(\beta)$, determined by $\theta(\alpha) = \beta$. Since E/F is a finite normal extension, by Theorem 7.5, E is the splitting field over F of some $f(x) \in F[x]$. We can also view E as the splitting field of $f(x)$ over both $F(\alpha)$ and $F(\beta)$. Thus by Theorem 4.4, there exists an isomorphism $\psi : E \rightarrow E$ extending θ , and so $\psi \in \text{Aut}_F(E)$ and $\psi(\alpha) = \beta$. ■

Example

The complex numbers $\sqrt[3]{2}$, $\sqrt[3]{2}\zeta_3$, $\sqrt[3]{2}\zeta_3^2$ are all conjugates over $\mathbb{Q}$ since they are roots of the irreducible polynomial $x^3 - 2 \in \mathbb{Q}[x]$.

Definition

A normal closure of a finite extension E/F is a finite normal extension N/F satisfying the following properties:

- (1) E is a subfield of N .
- (2) Let L be an intermediate field of N/E . Then if L/F is normal, then $L = N$.

Example

The normal closure of $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is $\mathbb{Q}(\sqrt[3]{2}, \zeta_3)/\mathbb{Q}$.

Theorem 7.8

Every finite extension E/F has a normal closure N/F which is unique up to E -isomorphism.

Proof:

Write $E = F(\alpha_1, \dots, \alpha_n)$. For existence, let $p_i(x)$ be the minimal polynomial of α_i over F for each $1 \leq i \leq n$. Write $f(x) = p_1(x) \dots p_n(x)$, and let N/E be the splitting field of $f(x)$ over E . Since $\alpha_1, \dots, \alpha_n$ are roots of $f(x)$, N is also the splitting field of $f(x)$ over F , so by Theorem 7.5, N/F is normal. Let $L \subseteq N$ be a subfield containing E . Then L contains all α_i 's. If L is normal over F , this implies each $p_i(x)$ splits over L , so by minimality of the splitting field, $N \subseteq L$, so $N = L$. Let N_1/F be another normal closure of E/F . Since N_1 is normal over F and contains all α_i , N_1 must contain a splitting field $\tilde{N}$ of $f(x)$ over F . Since N is a splitting field of $f(x)$ over F , by Corollary 4.5, $\tilde{N}$ is E -isomorphic to N . Since $\tilde{N}$ is a splitting field of $f(x)$ over F , it is normal over F by Theorem 7.5. Thus by definition of normal closure, $\tilde{N} = N_1$, and so N_1 is E -isomorphic to N . ■

Chapter 8: Galois Correspondence

8.1 Galois Extensions

Recall that for a finite extension E/F , we have proved that E is the splitting field of some $f(x) \in F[x]$ if and only if E/F is normal, and if E is the splitting field of some polynomial whose irreducible factors are separable $f(x) \in F[x]$, then E/F is separable (Theorems 7.5 and 7.1). We note that if we know a priori that E is the splitting field of some $f(x) \in F[x]$, then the latter statement becomes an ‘if and only if’.

Definition

An algebraic extension E/F is Galois if it is normal and separable. If E/F is a Galois extension, the Galois group of E/F is defined to be the automorphism group $\text{Gal}_F(E) = \text{Aut}_F(E)$.

Remark

- (1) Combining Theorems 7.1 and 7.5, a finite Galois extension E/F is equivalent to saying E is the splitting field of a polynomial $f(x) \in F[x]$ whose irreducible factors are separable.
- (2) If E/F is a finite Galois extension, by Theorem 6.9, $|\text{Gal}_F(E)| = [E : F]$.
- (3) If E/F is the splitting field of a separable polynomial $f(x) \in F[x]$ with $\deg(f) = n$, by Theorem 6.10, $\text{Gal}_F(E)$ is a subgroup of S_n .

Example

Let E be the splitting field of $f(x) = (x^2 - 2)(x^2 - 3)(x^2 - 5) \in \mathbb{Q}[x]$. Then $E = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$, and it's not hard to show $[E : \mathbb{Q}] = 8$. Then for $\psi \in \text{Gal}_{\mathbb{Q}}(E)$, we have $\psi(\sqrt{2}) = \pm\sqrt{2}$, $\psi(\sqrt{3}) = \pm\sqrt{3}$, and $\psi(\sqrt{5}) = \pm\sqrt{5}$, so 2 choices for each root. Since $|\text{Gal}_{\mathbb{Q}}(E)| = [E : \mathbb{Q}] = 8$, we have $\text{Gal}_{\mathbb{Q}}(E) \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$.

Definition

Let $t_1, \dots, t_n$ be variables. We define the elementary symmetric functions in $t_1, \dots, t_n$ as $s_1 = t_1 + \dots + t_n$, $s_2 = \sum_{1 \leq i < j \leq n} t_i t_j$, $s_n = t_1 \dots t_n$. Generally, and $s_k = \sum_{1 \leq i_1 < \dots < i_k \leq n} t_{i_1} \dots t_{i_k}$ for $1 \leq k \leq n$. Then it

follows that $f(x) = (x - t_1) \dots (x - t_n) = x^n - s_1 x^{n-1} + s_2 x^{n-2} + \dots + (-1)^n s_n$.

Theorem 8.1 (Artin)

Let E be a field and G a finite subgroup of $\text{Aut}(E)$. Let $E^G = \{\alpha \in E : \psi(\alpha) = \alpha, \forall \psi \in G\}$. Then E/E^G is a finite Galois extension and $\text{Gal}_{E^G}(E) = G$. In particular, $[E : E^G] = |G|$.

Proof:

Let $n = |G|$ and let $F = E^G$. For $\alpha \in E$, consider the G -orbit of α , $\{\psi(\alpha) : \psi \in G\} = \{\alpha = \alpha_1, \dots, \alpha_m\}$ for distinct α_i 's. Note that $m \leq n$, because we might have repeated roots. Let $f(x) = (x - \alpha_1) \dots (x - \alpha_m)$. Because they are all in the same G -orbit, for any $\psi \in G$, ψ permutes the roots $\{\alpha_1, \dots, \alpha_m\}$, so $\psi(f(x)) = f(x)$, so the coefficients of $f(x)$ are fixed by all $\psi \in G$ (alternatively, their coefficients are symmetric in the α_i 's). Thus $f(x) \in E^G[x] = F[x]$. To show $f(x)$ is the minimal polynomial of α , consider a factor $g(x) \in F[x]$ of $f(x)$. Without loss of generality, we can write $g(x) = (x - \alpha_1) \dots (x - \alpha_\ell)$ with $\ell \leq m$. If $\ell < m$, since all α_i 's ($1 \leq i \leq m$) are in the G -orbit of α , there exist $\psi \in G$ such that $\{\alpha_1, \dots, \alpha_\ell\} \neq \{\psi(\alpha_1), \dots, \psi(\alpha_\ell)\}$. It follows that $\psi(g(x)) \neq g(x)$, so some coefficient of $g(x)$ must not be fixed by ψ , so $g(x) \notin F[x]$. Thus any proper factor of $f(x)$ is not in $F[x]$, so $f(x)$ is irreducible over F , so it follows that $f(x)$ is the minimal polynomial of α over F . Thus, we see the minimal polynomial of an arbitrary $\alpha \in E$ over F splits over E , so E/F is normal. Since this minimal polynomial $f(x) \in F[x]$ is separable and α was arbitrary, this shows E/F is separable, so all-in-all E/F is a Galois extension. We claim $[E : F] \leq n$. If this claim holds, then E/F will be a finite Galois extension, so E will be the splitting field of polynomial over F whose irreducible factors are separable, and since $E^G = F$ (i.e. G fixes F), G is a subgroup of $\text{Gal}_F(E)$. But then we have by Theorem 6.9, $n = |G| \leq |\text{Gal}_F(E)| = [E : F] \leq n$, so $[E : F] = n$ and $\text{Gal}_F(E) = G$, which is the result. This leaves us to prove the claim $[E : F] \leq n$. Suppose $[E : F] > n$, allowing us to choose $\beta_1, \dots, \beta_n, \beta_{n+1} \in E$ which are linearly independent over F . Accordingly, consider the linear system

$$\psi(\beta_1)v_1 + \dots + \psi(\beta_{n+1})v_{n+1} = 0, \quad v_i \in E$$

for each $\psi \in G$, which is a system of $n = |G|$ linear equations in $n + 1$ variables $v_1, \dots, v_{n+1}$. Because the system is underdetermined, there is a nonzero solution $(\gamma_1, \dots, \gamma_{n+1})$, and we can choose a solution which has the minimal number, say r , of nonzero coordinates. We have $r \geq 2$, since if only one coordinate is nonzero, then this implies $\psi(\beta_i)v_i = 0$ for β_i, v_i nonzero which is impossible (we need things to cancel). Relabelling our variables, we may assume $\gamma_1, \dots, \gamma_r \neq 0$, but $\gamma_{r+1}, \dots, \gamma_n = 0$. Therefore,

$$\psi(\beta_1)\gamma_1 + \dots + \psi(\beta_r)\gamma_r = 0$$

for all $\psi \in G$. By dividing by γ_r , we may assume that $\gamma_r = 1$. Since $(\beta_1, \dots, \beta_r)$ are independent over F , and $\beta_1\gamma_1 + \dots + \beta_r\gamma_r = 0$ (take ψ to be the identity), we have at least one $\gamma_i \notin F$. Since $r \geq 2$ (so $i \neq r$ since $\gamma_r = 1 \in F$), relabelling if necessary, we can assume $\gamma_1 \notin F$. Thus, γ_1 is not fixed by G , so choose $\phi \in G$ such that $\phi(\gamma_1) \neq \gamma_1$. Applying ϕ to our above equation, we get

$$(\phi \circ \psi)(\beta_1)\phi(\gamma_1) + \dots + (\phi \circ \psi)(\beta_r)\phi(\gamma_r) = 0$$

for all $\psi \in G$, but since ϕ is an automorphism, $\phi \circ \psi$ runs through all elements of G as ψ does, so this shows

$$\psi(\beta_1)\phi(\gamma_1) + \dots + \psi(\beta_r)\phi(\gamma_r) = 0$$

Notice that since $\gamma_r = 1$, we have $\phi(\gamma_r) = 1$. Thus subtracting this equation from our earlier one, we have

$$\psi(\beta_1)(\gamma_1 - \phi(\gamma_1)) + \dots + \psi(\beta_{r-1})(\gamma_{r-1} - \phi(\gamma_{r-1})) + \psi(\beta_r)(\gamma_r - \phi(\gamma_r)) = 0$$

But $\gamma_1 \neq \phi(\gamma_1)$ by construction, so this is a nontrivial linear combination, but looking at the last coordinate, we have $\gamma_r - \phi(\gamma_r) = 1 - 1 = 0$, so this shows $(\gamma_1 - \phi(\gamma_1), \dots, \gamma_{r-1} - \phi(\gamma_{r-1}), 0, \dots, 0)$ is a nonzero solution to our original system with at most $r - 1$ nonzero coordinates, contradicting minimality of r , proving the claim and thus the theorem. ■

Remark

Let E be a field and G a finite subgroup of $\text{Aut}(E)$. For $\alpha \in E$, let $\{\alpha = \alpha_1, \dots, \alpha_m\}$ be the G -orbit of α . Then we see from the proof of Theorem 8.1 that the minimal polynomial of α over E^G is $(x - \alpha_1) \dots (x - \alpha_m) \in E^G[x]$.

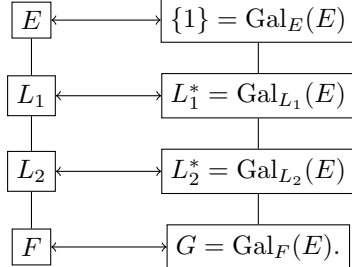
Example

Let $E = F(t_1, \dots, t_n)$ be the function field in n variables $t_1, \dots, t_n$ over a field F . Consider the symmetric group $G := S_n$ as the subgroup of $\text{Aut}(E)$ which permutes the variables $t_1, \dots, t_n$ and fixes F . We are interested in finding E^G . From the proof of Theorem 8.1, the coefficients of the minimal polynomial of t_1 lie in E^G , so by considering such a polynomial, we can get some hints about E^G . The G -orbit of t_1 is $\{t_1, \dots, t_n\}$. By the above remark, $f(x) = (x - t_1) \dots (x - t_n)$ is the minimal polynomial of t_1 over E^G . Let $s_1, \dots, s_n$ be the elementary symmetric functions of $t_1, \dots, t_n$. Then $f(x) = x^n - s_1x^{n-1} + s_2x^{n-2} + \dots + (-1)^n s_n \in F(s_1, \dots, s_n)[x]$. We claim $F(s_1, \dots, s_n) = E^G$. Indeed, because L is generated by symmetric functions, L is fixed by G and so $L \subseteq E^G$. Conversely, note that E is the splitting field of $f(x)$ over L . Since $\deg(f) = n$, by Theorem 4.6, we have $[E : F(s_1, \dots, s_n)] \leq n!$. On the other hand, by Theorem 8.1, $n! = |G| = [E : E^G] \leq [E : F(s_1, \dots, s_n)] = n!$, so $E^G = F(s_1, \dots, s_n)$.

8.2 The Fundamental Theorem

Theorem 8.2 (Fundamental Theorem of Galois Theory)

Let E/F be a finite Galois extension and $G = \text{Gal}_F(E)$. Then there is an order-reversing bijection between the intermediate fields of E/F and the subgroups of G . More precisely, let $\text{Int}(E/F)$ denote the set of intermediate fields of E/F and $\text{Sub}(G)$ the set of subgroups of G , then the map $\text{Int}(E/F) \rightarrow \text{Sub}(G)$ defined by $L \mapsto L^* := \text{Gal}_L(E)$ has inverse $\text{Sub}(G) \rightarrow \text{Int}(E/F)$ defined by $H \mapsto H^* := E^H$, and these maps reverse the inclusion relation: for $L_1, L_2 \in \text{Int}(E/F)$ with $L_2 \subseteq L_1$ and $H_1, H_2 \in \text{Sub}(G)$ with $H_2 \subseteq H_1$, we have $[L_1 : L_2] = [L_2^* : L_1^*]$ and $[H_1 : H_2] = [H_2^* : H_1^*]$.



Proof:

Let $L \in \text{Int}(E/F)$ and $H \in \text{Sub}(G)$. We recall Theorem 6.11, which states that if $G_1 = \text{Gal}_{F_1}(E_1)$, then $E_1^{G_1} = F_1$. Thus $(L^*)^* = (\text{Gal}_L(E))^* = E^{\text{Gal}_L(E)} = L$. Also, Theorem 8.1 states if $G_1 \subseteq \text{Aut}(E_1)$, then $\text{Gal}_{E_1^{G_1}}(E_1) = G_1$. Thus $(H^*)^* = (E^H)^* = \text{Gal}_{E^H}(E) = H$. Therefore, we see $L \mapsto L^*$ and $H \mapsto H^*$ are inverses of each other. Let $L_1, L_2 \in \text{Int}(E/F)$. Since E/F is the splitting field of some $f(x) \in F[x]$ whose irreducible factors are separable, E/L_1 and E/L_2 are also Galois extensions since E is the splitting field of $f(x)$ over L_1 and L_2 respectively. If $L_2 \subseteq L_1$, then the automorphisms of E that fix L_1 also fix $L_2 \subseteq L_1$, and so $\text{Gal}_{L_1}(E) \subseteq \text{Gal}_{L_2}(E)$, i.e. $L_1^* \subseteq L_2^*$. Moreover, $[L_1 : L_2] = \frac{[E:L_2]}{[E:L_1]} = \frac{|\text{Gal}_{L_2}(E)|}{|\text{Gal}_{L_1}(E)|} = \frac{|L_2^*|}{|L_1^*|} = [L_2^* : L_1^*]$, using Theorems 3.1 (first equality) and 6.9 (second equality). Let $H_1, H_2 \in \text{Sub}(G)$. If $H_2 \subseteq H_1$, then the subfield of E fixed by H_1 is also fixed by H_2 , so $E^{H_1} \subseteq E^{H_2}$, i.e. $H_1^* \subseteq H_2^*$. Moreover, $[H_1 : H_2] = \frac{|H_1|}{|H_2|} = \frac{|\text{Gal}_{E^{H_1}}(E)|}{|\text{Gal}_{E^{H_2}}(E)|} = \frac{[E:E^{H_1}]}{[E:E^{H_2}]} = [E^{H_2} : E^{H_1}] = [H_2^* : H_1^*]$. ■

Example

Let's return to the very first question we asked in the intro of the course. Consider $E = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ and $\mathbb{Q}$. What are the intermediate fields of $E/\mathbb{Q}$? Since $\text{Gal}_{\mathbb{Q}}(E) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ has only finitely many subgroups, there are only finitely many intermediate fields between E and $\mathbb{Q}$.

Example

We have seen that if E/F is a finite Galois extension and $L \in \text{Int}(E/F)$, then L/F is not always Galois. For example, $E = \mathbb{Q}(\sqrt[3]{2}, \zeta_3)$, $L = \mathbb{Q}(\sqrt[3]{2})$, $F = \mathbb{Q}$. Can we use the fundamental theorem to determine when L/F is Galois? From the theorem, if L/F is Galois, it corresponds to the group G/L^* , which is well-defined only if L^* is normal in G .

Proposition 8.3

Let E/F be a finite Galois extension with $G = \text{Gal}_F(E)$. Let L be an intermediate field. Then, for $\psi \in G$,

$$\text{Gal}_{\psi(L)}(E) = \psi \text{Gal}_L(E) \psi^{-1}$$

Proof:

For $\alpha \in \psi(L)$, thus have $\psi^{-1}(\alpha) \in L$, so if $\phi \in \text{Gal}_L(E)$, we have $\phi\psi^{-1}(\alpha) = \psi^{-1}(\alpha)$ since ϕ fixes L , thus $\psi\phi\psi^{-1}(\alpha) = \alpha$. Thus $\psi\phi\psi^{-1}$ fixes $\psi(L)$, so $\psi\phi\psi^{-1} \in \text{Gal}_{\psi(L)}(E)$, so $\psi \text{Gal}_L(E) \psi^{-1} \subseteq \text{Gal}_{\psi(L)}(E)$. Since ψ is an automorphism, $|\psi \text{Gal}_L(E) \psi^{-1}| = |\text{Gal}_L(E)| = [E : L] = [E : \psi(L)] = |\text{Gal}_{\psi(L)}(E)|$, where the last equality is Theorem 8.2, and so for size considerations, the inclusion $\psi \text{Gal}_L(E) \psi^{-1} \subseteq \text{Gal}_{\psi(L)}(E)$ is an equality. ■

Theorem 8.4

Let $E/F, L, L^*$ be defined as in Theorem 8.2. Then L/F is a Galois extension if and only if L^* is a normal subgroup of G . In this case, $\text{Gal}_F(L) \cong G/L^*$.

Proof:

We have that L/F is normal if and only if $\psi(L) = L$ for all $\psi \in \text{Gal}_F(E) = G$, if and only if $\text{Gal}_{\psi(L)}(E) = \text{Gal}_L(E) = L^*$ for all $\psi \in G$, which by Proposition 8.3 tells us holds if and only if $\psi L^* \psi^{-1} = L^*$ for all $\psi \in G$, i.e. if and only if L^* is normal in G . Because L/F is Galois, the restriction map $G = \text{Gal}_F(E) \rightarrow \text{Gal}_F(L)$, $\psi \mapsto \psi|_L$ is well-defined (see A8Q2). Moreover, it is surjective and its kernel is $\text{Gal}_L(E) = L^*$, since its kernel is the maps that fix L , so by the First Isomorphism Theorem, $\text{Gal}_F(L) \cong G/L^*$. ■

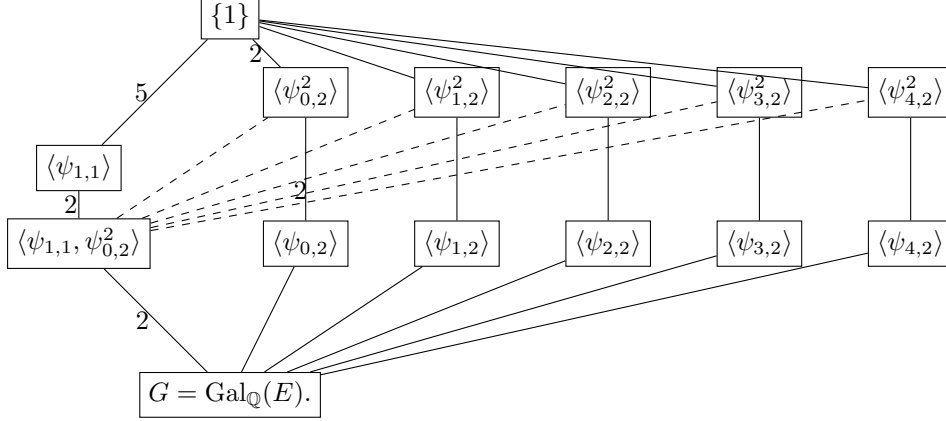
Example

For a prime p , let $q = p^n$. We have seen on Assignment 4 the Frobenius automorphism of $\mathbb{F}_q$ defined by $\sigma_p : \mathbb{F}_q \rightarrow \mathbb{F}_q$, $\alpha \mapsto \alpha^p$, where for $\alpha \in \mathbb{F}_q$, we have $\sigma_p^n(\alpha) = \alpha^{p^n} = \alpha$. For $1 \leq m < n$, we have $\sigma_p^m(\alpha) = \alpha^{p^m}$. Since the polynomial $x^{p^m} - x$ has at most p^m roots in $\mathbb{F}_q$, there exist $\alpha \in E$ such that $\alpha^{p^m} - \alpha \neq 0$. Thus $\sigma_p^m \neq 1$, so σ_p has order n . Let $G = \text{Gal}_{\mathbb{F}_p}(\mathbb{F}_q)$. It follows that $n = |\langle \sigma_p \rangle| \leq |G| = [\mathbb{F}_q : \mathbb{F}_p] = n$, and so $\langle \sigma_p \rangle = G$, a cyclic group of order n . Consider a subgroup of H of G of order d . Thus $d \mid n$ and $[G : H] = n/d$. By Theorem 8.2, $n/d = [G : H] = [H^* : G^*] = [\mathbb{F}_q^H : \mathbb{F}_q^G] = [\mathbb{F}_q^H : \mathbb{F}_p]$, so $H^* = \mathbb{F}_q^H$, which has degree n/d and so $H^* = \mathbb{F}_{p^{n/d}}$.

Example

Let E be the splitting field of $x^5 - 7$ over $\mathbb{Q}$ in $\mathbb{C}$. Then $E = \mathbb{Q}(\alpha, \zeta_5)$ with $\alpha = \sqrt[5]{7}$ and $\zeta_5 = e^{2\pi i/5}$. The minimal polynomials of α and ζ_5 over $\mathbb{Q}$ are $x^5 - 7$ and $x^4 + x^3 + x^2 + x + 1$ respectively. Then $[E : \mathbb{Q}] = 20$, and hence $G = \text{Gal}_{\mathbb{Q}}(E)$ is (isomorphic to) a subgroup of S_5 of order 20 by Theorem 6.10. For $\psi \in G$, its action is determined by $\psi(\alpha)$ and $\psi(\zeta_5)$. Accordingly, write $\psi = \psi_{k,s}$ if $\psi(\alpha) = \alpha\zeta_5^k$, $k \in \mathbb{Z}_5$ and $\psi(\zeta_5) = \zeta_5^s$, $s \in \mathbb{Z}_5^*$. Define $\sigma = \psi_{1,1}$ and $\tau = \psi_{0,2}$. Then $\tau\sigma = \sigma^2\tau$, and so we have the presentation $G = \langle \sigma, \tau : \sigma^5 = 1 = \tau^4, \tau\sigma = \sigma^2\tau \rangle$, in particular it isn't abelian. Since $|G| = 20$, by Lagrange's Theorem, the possible subgroups of G are of order 1, 2, 4, 5, 10, 20. We have $|G| = 20 = 2^2 \cdot 5$. Letting n_p be the number of p -Sylow subgroups of G , by Sylow's third theorem, $n_5 \mid 4$ and $n_5 \equiv 1 \pmod{4}$, which forces $n_5 = 1$. Also, $n_2 \mid 5$ and $n_2 \equiv 1 \pmod{5}$, hence $n_2 = 1$ or $n_2 = 5$. If $n_2 = 1$, then both p -Sylow subgroups are normal, so by the Chinese remainder theorem, $G \cong \mathbb{Z}_4 \times \mathbb{Z}_5$, which contradicts G being nonabelian. Therefore, there are 5 2-Sylow subgroups. Since $\tau \in G$ is of order 4, so $\langle \tau \rangle$ is a 2-Sylow subgroup of G , and by Sylow's second theorem, all other 2-Sylow subgroups of G are conjugates of $\langle \tau \rangle$. But elements of G are of the form $\sigma^a \tau^b$, so conjugates of τ are of the form $\sigma^a \tau^b \tau^{-b} \sigma^{-a} = \sigma^a \tau \sigma^{-a}$.

Using the relation that $\tau\sigma = \sigma^2\tau$ and $\sigma^{-1} = \sigma^4$, we have $\langle\sigma^4\tau\sigma^{-4}\rangle = \langle\sigma^{-1}\tau\sigma\rangle = \langle\sigma\tau\rangle = \langle\psi_{1,2}\rangle$. Using the same argument, we have that the 2-Sylow subgroups of G are $\langle\psi_{0,2}\rangle, \langle\psi_{1,2}\rangle, \langle\psi_{2,2}\rangle, \langle\psi_{3,2}\rangle, \langle\psi_{4,2}\rangle$. Moreover, since a 2-subgroup of G is contained in a 2-Sylow subgroup, the groups of order 2 of G are $\langle\psi_{0,2}^2\rangle, \langle\psi_{1,2}^2\rangle, \langle\psi_{2,2}^2\rangle, \langle\psi_{3,2}^2\rangle, \langle\psi_{4,2}^2\rangle$. Thus, we explicitly know all the subgroups of G of order 1, 2, 4, 5, 20, leaving 10 to figure out. Let H be a subgroup of G of order 10. This must contain the unique 5-Sylow subgroup $\langle\sigma\rangle$, so $\sigma^a\tau^b \in H$ if and only if $\tau^b \in H$. But the only element of the form τ^b which is of order 2 is τ^2 , hence $H = \langle\sigma, \tau^2\rangle$. Therefore, we know all the subgroups of G , so we can explicitly write out the Galois correspondence:



For an intermediate field L of $E/\mathbb{Q}$, we consider $L^* = \text{Gal}_L(E)$. For example, for $\mathbb{Q}(\zeta_5)$, note that $\psi_{1,1}(\zeta_5) = \zeta_5$. Thus $\mathbb{Q}(\zeta_5)^* \supseteq \langle\psi_{1,1}\rangle$. Since $|\langle\psi_{1,1}\rangle| = [\langle\psi_{1,1}\rangle : \{1\}] = 5$ and $5 = [E : \mathbb{Q}(\zeta_5)] = [\mathbb{Q}(\zeta_5)^* : \{1\}]$, we have $\mathbb{Q}(\zeta_5)^* = \langle\psi_{1,1}\rangle$.

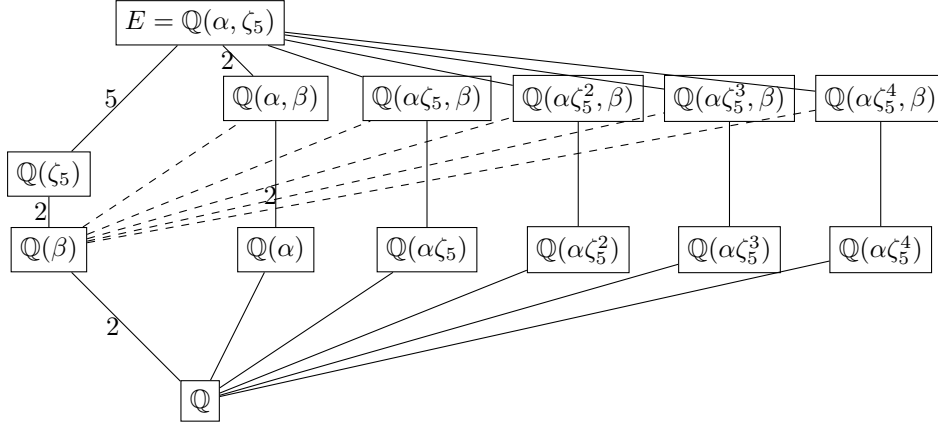
Also, $\psi_{1,2}(\alpha\zeta_5^r) = \alpha\zeta_5^{2r+1}$, so if $\psi_{1,2}$ fixed $\alpha\zeta_5^r$, then $r \equiv 2r+1 \pmod{5}$, i.e. $r \equiv 4 \pmod{5}$. Thus $\mathbb{Q}(\alpha\zeta_5^4)^* \supseteq \langle\psi_{1,2}\rangle$, and this is an equality for the same index considerations as before. Using the same argument, we can get $\langle\psi_{r,2}\rangle^*$ for $r = 0, 1, 2, 3, 4$. Let $\beta = \zeta_5 + \zeta_5^{-1} \in \mathbb{R}$. We have

$$\beta^2 + \beta - 1 = \zeta_5^2 + 2 + \zeta_5^{-2} + \zeta_5 + \zeta_5^{-1} - 1$$

but $\zeta_5^{-2} = \zeta_5^3$, and $\zeta_5^{-1} = \zeta_5^4$, so

$$\beta^2 + \beta - 1 = 1 + \zeta_5 + \zeta_5^2 + \zeta_5^3 + \zeta_5^4 = 0$$

Thus β is a root of the quadratic polynomial $x^2 + x - 1$, which we can check has no rational roots, so $[\mathbb{Q}(\beta) : \mathbb{Q}] = 2$, and similarly $[\mathbb{Q}(\alpha, \beta) : \mathbb{Q}(\alpha)] = 2$. We thus have the corresponding diagram of the intermediate fields of $E/\mathbb{Q}$:



Chapter 9: Cyclic Extensions

Definition

A Galois extension E/F is called cyclic, abelian, or solvable if $\text{Gal}_F(E)$ has the corresponding property.

Lemma 9.1 (Dedekind's Lemma)

Let $\psi_i : L \rightarrow K$ be a distinct nonzero homomorphism for $1 \leq i \leq n$. If for some $c_1, \dots, c_n \in K$, we have $c_1\psi_1(\alpha) + \dots + c_n\psi_n(\alpha) = 0$ for all $\alpha \in L$, then $c_1 = \dots = c_n = 0$.

Proof:

Suppose for the sake of contradiction that there exists some $c_1, \dots, c_n \in K$ not all zero such that $c_1\psi_1(\alpha) + \dots + c_n\psi_n(\alpha) = 0$ for all $\alpha \in L$. Let m be the minimal positive integer such that $c_1\psi_1(\alpha) + \dots + c_m\psi_m(\alpha) = 0$ for all $\alpha \in L$. Reordering if necessary, by minimality of m , we have $c_i \neq 0$ for each $1 \leq i \leq m$, and since $\psi_1 \neq \psi_2$, we can choose $\beta \in L$ such that $\psi_1(\beta) \neq \psi_2(\beta)$. Without loss of generality, suppose $\psi_1(\beta) \neq 0$. Then by the above (since it holds for all α), we have $c_1\psi_1(\alpha\beta) + \dots + c_m\psi_m(\alpha\beta) = 0$ for all $\alpha \in L$. Since $\psi_1(\beta) \neq 0$, by dividing this by $\psi_1(\beta)$, we get $c_1\psi_1(\alpha) + c_2\psi_2(\alpha)\psi_2(\beta)/\psi_1(\beta) + \dots + c_m\psi_m(\alpha)\psi_m(\beta)/\psi_1(\beta) = 0$. Subtracting this from our original equation, we get

$$c_2(1 - \frac{\psi_2(\beta)}{\psi_1(\beta)})\psi_2(\alpha) + \dots + c_m(1 - \frac{\psi_m(\beta)}{\psi_1(\beta)})\psi_m(\alpha) = 0$$

As $c_2(1 - \frac{\psi_2(\beta)}{\psi_1(\beta)}) \neq 0$ since $\psi_2(\beta) \neq \psi_1(\beta)$, this contradicts minimality of m , proving the result. ■

Theorem 9.2

Let F be a field and $n \in \mathbb{N}$. Suppose $\text{ch}(F) = 0$ or $\text{ch}(F) = p$ with $p \nmid n$, and that $x^n - 1$ splits over F . Then:

- (1) If a Galois extension E/F is cyclic of degree n , then $E = F(\alpha)$ for some $\alpha \in E$ such that $\alpha^n \in F$. In particular, $x^n - \alpha^n$ is the minimal polynomial of α over F .
- (2) If $E = F(\alpha)$ for some $\alpha \in E$ satisfying $\alpha^n \in F$, then E/F is a cyclic extension of degree d with $d \mid n$ and $\alpha^d \in F$. In particular, $x^d - \alpha^d$ is the minimal polynomial of α over F .

Proof:

Let ζ_n be a primitive n th root of unity, i.e. $\zeta_n^n = 1$ and $\zeta_n^d \neq 1$ for all $1 \leq d < n$. Note that one exists in F since $x^n - 1$ splits over F , and since $\text{ch}(F) = 0$ or $\text{ch}(F) = p$ with $p \nmid n$, the polynomial $x^n - 1$ is separable (its derivative is nx^{n-1} which is coprime under our assumptions about $\text{ch}(F)$). Thus $1, \zeta_n, \dots, \zeta_n^{n-1}$ are all distinct.

(1) Let $G = \text{Gal}_F(E) = \langle \psi \rangle \cong C_n$, the cyclic group of order n . Apply Dedekind's lemma to $K = L = E$, where the ψ_i 's are all elements of G , and $c_1 = 1, c_2 = \zeta_n^{-1}, \dots, c_n = \zeta_n^{-(n-1)}$. Since $c_i \neq 0$ for all i , there exists $u \in E$ such that $\alpha = u + \zeta_n^{-1}\psi(u) + \dots + \zeta_n^{-(n-1)}\psi^{n-1}(u) \neq 0$, using the fact G is cyclic. But now note that $1(\alpha) = \alpha$, $\psi(\alpha) = \psi(u) + \zeta_n^{-1} + \zeta_n^{-1}\psi^2(u) + \dots + \zeta_n^{-(n-1)}\psi^n(u) = \alpha\zeta_n$ (everything from the above equation for α is shifted by one), and similarly, $\psi^2(\alpha) = \alpha\zeta_n^2, \dots, \psi^{n-1}(\alpha) = \alpha\zeta_n^{n-1}$. Thus, we see $\alpha, \alpha\zeta_n, \dots, \alpha\zeta_n^{n-1}$ are conjugate to each other, so they have the same minimal polynomial over F , say $p(x)$. Moreover, $\alpha, \alpha\zeta_n, \dots, \alpha\zeta_n^{n-1}$ are all distinct, so $\deg(p) = n$. Also, since $p(x) \in F[x]$, we have $p(0) = \pm\alpha(\alpha\zeta_n) \dots (\alpha\zeta_n^{n-1}) = \pm\alpha^n\zeta_n^{n(n-1)/2}$ (the constant term of $p(x)$). Since $\zeta_n \in F$, this shows $\alpha^n \in F$. Since α is a root of $x^n - \alpha^n \in F[x]$ and $\deg(p) = n$ and $p(x)$ is the minimal polynomial of α , by minimality we must have $p(x) = x^n - \alpha^n$. Moreover, since $F(\alpha) \subseteq E$ and $[F(\alpha) : F] = \deg(p) = n = [E : F]$, we have $E = F(\alpha)$.

(2) Suppose $E = F(\alpha)$ for some $\alpha \in F$ satisfying $\alpha^n \in F$. Let $p(x) \in F[x]$ be the minimal polynomial of α over F . Since α^n is a root of $x^n - \alpha^n \in F[x]$, $p(x) \mid x^n - \alpha^n$. Thus the roots of $p(x)$ are of the form $\alpha\zeta_n^i$ for some i by the same argument as above, so we have $p(0) = \pm\alpha^d\zeta_n^k$ for some $k \in \mathbb{Z}$ and $d = \deg(p)$. Since $p(0) \in F$ and $\zeta_n \in F$, we have $\alpha^d \in F$. Since $x^d - \alpha^d \in F[x]$ has α as a root, $p(x) \mid x^d - \alpha^d$, which since $\deg(p) = d$, shows $p(x) = x^d - \alpha^d$. We claim $d \mid n$. Indeed, if not, long divide and write $n = qd + r$ for $q \in \mathbb{Z}$ and $0 < r < d$. Since $\alpha^d, \alpha^n \in F$, we get $\alpha^r = \alpha^{n-qd} \in F$, so α is a root of $x^r - \alpha^r \in F[x]$, which is of lower degree than $p(x)$ since $r < d$, contradicting minimality of $p(x)$, so we must have $d \mid n$. Thus write $n = md$. Since $p(x) = x^d - \alpha^d$, the roots of $p(x)$ are $\alpha, \alpha\zeta_n^m, \alpha\zeta_n^{2m}, \dots, \alpha\zeta_n^{(d-1)m}$. Since

$\zeta_n \in F$, $E = F(\alpha)$ is the splitting field of the separable polynomial $p(x)$ over F , and so is Galois over F . Now, pick $\psi \in \text{Gal}_E(F)$ satisfying $\psi(\alpha) = \alpha\zeta_n$. Then by the description of the roots above, since $\zeta_n \in F$, one has $G = \langle \psi \rangle \cong C_d$. Thus E/F is a cyclic extension of degree d . ■

Theorem 9.3

Let F be a field with $\text{ch}(F) = p$, where p is a prime.

- (1) If $x^p - x - a \in F[x]$ is irreducible, then its splitting field E/F is a cyclic extension of degree p .
- (2) If E/F is a cyclic extension of degree p , then E/F is the splitting field of some irreducible polynomial $x^p - x - a \in F[x]$.

Proof:

(1) Let $f(x) = x^p - x - a$ and suppose α is a root of $f(x)$. Then since $\text{ch}(F) = p$, $f(\alpha + 1) = (\alpha + 1)^p - (\alpha + 1) - a = \alpha^p + 1 - \alpha - 1 - a = \alpha^p - \alpha - a = f(\alpha) = 0$. Therefore, $\alpha + 1$ is also a root of $f(x)$, so by induction, $\alpha + 1, \alpha + 2, \dots, \alpha + (p - 1)$ are also roots of $f(x)$. Since $f(x)$ has at most p distinct roots, and we have found p of them, it follows that we have found all the roots of $f(x)$. Therefore, $E = F(\alpha, \alpha + 1, \dots, \alpha + (p - 1)) = F(\alpha)$, and $[E : F] = \deg f = p$. Therefore, $\text{Gal}_F(E)$ is of prime order, and so is isomorphic to the cyclic group of order p , where its generator is the automorphism determined by $\alpha \mapsto \alpha + 1$. Thus E/F is cyclic of degree p .

(2) Let $G = \text{Gal}_F(E) = \langle \psi \rangle \cong C_p$. Apply Dedekind's Lemma 9.1 to $K = L = E$, ψ_i all elements of G and $c_1 = \dots = c_p = 1$, getting that there exists $v \in E$ such that $\beta := v + \psi(v) + \psi^2(v) + \dots + \psi^{p-1}(v) \neq 0$. Then, notice that applying ψ^i to β for all $0 \leq i \leq p - 1$ just shifts the above expression, so $\psi^i(\beta) = \beta$, so $\beta \in F$ since it is fixed by every element of $\text{Gal}_F(E)$. Let $u = v/\beta$. Since $\beta \in F$, we have

$$u + \psi(u) + \dots + \psi^{p-1}(u) = v/\beta + \psi(v/\beta) + \dots + \psi^{p-1}(v/\beta) = \frac{v + \psi(v) + \dots + \psi^{p-1}(v)}{\beta} = \beta/\beta = 1$$

Set $\alpha := -\psi(u) - 2\psi^2(u) - \dots - (p - 1)\psi^{p-1}(u)$. Then $\psi(\alpha) = -\psi^2(u) - 2\psi^3(u) - \dots - (p - 1)\psi^p(u)$, so $\psi(\alpha) - \alpha = \psi(u) + \psi^2(u) + \dots + \psi^{p-1}(u) + \psi^p(u) = 1$ by our above calculation (since $\psi^p(u) = u$), so $\psi(\alpha) = \alpha + 1$. Since $\text{ch}(F) = p$, we get $\psi(\alpha^p) = \psi(\alpha)^p = (\alpha + 1)^p = \alpha^p + 1$, so $\psi(\alpha^p - \alpha) = \psi(\alpha^p) - \psi(\alpha) = (\alpha^p + 1) - (\alpha + 1) = \alpha^p - \alpha$, so $\alpha^p - \alpha$ is fixed by ψ which generates $\text{Gal}_F(E)$, so $a := \alpha^p - \alpha \in F$ and α is a root of $x^p - x - a \in F[x]$. Since $[E : F] = p$, $[F(\alpha) : F]$ is a factor of p . Since $\alpha \notin F$ since it is not fixed by ψ , we get $[F(\alpha) : F] = p$ by primality, so it follows that $F(\alpha) = E$ and that $x^p - x - a \in F[x]$ is the minimal polynomial of α over F (we know all of its roots are the distinct $\alpha, \alpha + 1, \dots, \alpha + (p - 1)$ and that $\text{Gal}_F(E)$ permutes these). ■

Chapter 10: Solvability by Radicals

10.1 Radical Extensions

Definition

A finite extension E/F is radical if there exists a tower of fields $F = F_0 \subseteq F_1 \subseteq \dots \subseteq F_m = E$ such that $F_i = F_{i-1}(\alpha_i)$ where $\alpha_i \in F_i$ and $\alpha_i^{d_i} \in F_{i-1}$ for some $d_i \in \mathbb{N}$ ($1 \leq i \leq m$).

Lemma 10.1

If E/F is a finite separable radical extension, then its normal closure N/F is also radical.

Proof:

Since E/F is a finite separable extension, by the Primitive Element Theorem 7.4, $E = F(\beta)$ for some $\beta \in E$. Since E/F is a radical, there is a tower $F = F_0 \subseteq F_1 \subseteq \dots \subseteq F_m = E$ such that $F_i = F_{i-1}(\alpha_i)$ and $\alpha_i^{d_i} \in F_{i-1}$ for some $d_i \in \mathbb{N}$. Let $p(x) \in F[x]$ be the minimal polynomial of β , and let $\beta = \beta_1, \dots, \beta_n$ be roots of $p(x)$. By definition of the normal closure and theorem 7.5, $N = E(\beta_2, \dots, \beta_n) = F(\beta_1, \dots, \beta_n)$ and there is an F -isomorphism $\sigma_j : F(\beta) \rightarrow F(\beta_j)$ determined by $\beta \mapsto \beta_j$ for all $j = 2, \dots, n$. Since N can be viewed as the splitting field of $p(x)$ over $F(\beta)$ and $F(\beta_j)$ respectively, by Theorem 4.4, there exists an isomorphism $\psi_j : N \rightarrow N$ extending σ_j for

each $j = 2, \dots, n$. Thus $\psi_j \in \text{Gal}_F(N)$ and $\psi_j(\beta) = \beta_j$. Thus we have the following tower of fields $F = F_0 \subseteq F_1 \subseteq \dots \subseteq F_m = E = F(\beta_1) = F(\beta_1)\psi_2(F_0)$ since ψ_2 fixes $F_0 = F$, and we can continue this tower $F(\beta_1)\psi_2(F_0) \subseteq F(\beta_1)\psi_2(F_1) \subseteq \dots \subseteq F(\beta_1)\psi_2(F_m)$. But $F_m = E = F(\beta_1)$, so $\psi_2(F_m) = \psi_2(F(\beta_1)) = F(\beta_2)$ by definition of ψ_2 , so we get $F(\beta_1)\psi_2(F_m) = F(\beta_1, \beta_2) = F(\beta_1, \beta_2)\psi_3(F_0)$, and we can continue the tower the same way with ψ_3 this time, and so on up to ψ_n , repeating until we end the tower with $F(\beta_1, \dots, \beta_n) = N$. Note that since $F_i = F_{i-1}(\alpha_i)$ and $\alpha_i^{d_i} \in F_{i-1}$, we have $F(\beta_1, \dots, \beta_{j-1})\psi_j(F_i) = F(\beta_1, \dots, \beta_{j-1})\psi_j(F_{i-1}(\alpha_i)) = (F(\beta_1, \dots, \beta_{j-1})\psi_j(F_{i-1}))(\psi_j(\alpha_i))$ and for this $\psi_j(\alpha_i)$, $(\psi_j(\alpha_i))^{d_i} = \psi_j(\alpha_i^{d_i}) \in \psi_j(F_{i-1}) \subseteq F(\beta_1, \dots, \beta_{j-1})\psi_j(F_{i-1})$. Thus, we see our tower meets the definition to say N/F is radical. ■

Remark

In light of the above Lemma, to consider a finite separable radical extension, we could instead consider its normal closure, which is Galois.

Definition

Let F be a field and let $f(x) \in F[x]$. We say $f(x)$ is solvable by radicals if there exists a radical extension E/F such that $f(x)$ splits over E .

Remark

It is possible that $f(x) \in F[x]$ is solvable by radicals, but its splitting field is not a radical extension (see Assignment 10, the example is $x^3 - 3x + 1$). The idea is that the radicals you need to take to get the roots of $f(x)$ involve a field extension that isn't necessary for $f(x)$ to just split: in the given example, its roots are real but to solve it by radicals one needs complex numbers.

Remark

Recall that an expression involving only addition, subtraction, multiplication, division, and n th roots is a radical. Let F be a field and let $f(x) \in F[x]$ be separable. If $f(x)$ is solvable, by the definition of a radical extension, $f(x)$ has a radical root. Conversely, if $f(x)$ has a radical root, then it is in some radical extension E/F , and by Lemma 10.1, the radical closure N/F of E/F is radical. Since $f(x)$ splits over N , $f(x)$ is solvable by radicals.

10.2 Radical Solutions

Lemma 10.2

Let E/F be a field extension and let K, L be intermediate fields of E/F . Suppose that K/F is a finite Galois extension. Then KL is a finite Galois extension of L and $\text{Gal}_L(KL)$ is isomorphic to a subgroup of $\text{Gal}_F(K)$.

Proof:

Since K/F is a finite Galois extension, K is the splitting field of some $f(x) \in F[x]$ over F whose irreducible factors are separable. Since $F \subseteq L$, KL is the splitting field of $f(x)$ over L , thus Galois. Consider the map $\Gamma : \text{Gal}_L(KL) \rightarrow \text{Gal}_F(K)$ by $\psi \mapsto \psi|_K$. Because K/F and KL/L are Galois (see Assignment 8), this map is well-defined. Moreover, if $\psi|_K = 1_K$, then ψ is trivial on both K and L (the latter by definition), so it is trivial on KL , which shows Γ is injective. Thus $\text{Gal}_L(KL) \cong \text{im } \Gamma$, which is a subgroup of $\text{Gal}_F(K)$. ■

Definition

Let E/F be the splitting field of a polynomial $f(x) \in F[x]$ whose irreducible factors are separable. The Galois group of $f(x)$ is $\text{Gal}_F(E)$, denoted by $\text{Gal}(f)$.

Theorem 10.3

Let F be a field with $\text{ch}(F) = 0$ and $f(x) \neq 0 \in F[x]$. Then $f(x)$ is solvable by radicals if and only if its Galois group $\text{Gal}(f)$ is solvable.

Proof:

In the reverse direction, suppose $G = \text{Gal}(f)$ is solvable. Let E/F be the splitting field of $f(x)$ over F , and let $n = |G|$. Let L/E be the splitting field of $x^n - 1$ over E and $\zeta_n \in L$ a primitive n th root of unity. Set $K = F(\zeta_n)$. Then $L = E(\zeta_n) = KE$. Since E/F is a finite Galois extension, by Lemma 10.2, L/K is a finite Galois extension and $H = \text{Gal}_K(L)$ is isomorphic to a subgroup of G , and thus solvable by Theorem 6.3. Write $H = H_0 \supseteq H_1 \supseteq \cdots \supseteq H_m = \{1\}$ where H_i is normal in H_{i-1} and each quotient H_{i-1}/H_i is cyclic of order d_i . Since H is a subgroup of G , we have $d_i \mid n$ for each i by Lagrange. Let $K_i = H_i^* = L^{H_i}$. By Theorem 6.11, we have $\text{Gal}_{K_i}(L) = H_i$. This gives a tower of fields $K = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_m = L = E(\zeta_n)$. Since H_i is normal in H_{i+1} , by Theorem 8.4, K_i/K_{i-1} is Galois and $\text{Gal}_{K_{i-1}}(K_i) \cong H_{i-1}/H_i \cong C_{d_i}$. We have that $\zeta_n \in K_{i-1}$, so $\zeta_{d_i} = \zeta_n^{n/d_i} \in K_{i-1}$ since $d_i \mid n$, by Theorem 9.2, there exists $\alpha_i \in K_i$ such that $K_i = K_{i-1}(\alpha_i)$ with $\alpha_i^{d_i} \in K_{i-1}$. Moreover, $K_0 = K = F(\zeta_n)$ with $\zeta_n^n = 1 \in F$, which shows L/F is a radical extension. Since all roots of $f(x)$ are in E and thus in L , $f(x)$ is thus solvable by radicals.

Conversely, suppose $f(x)$ is solvable by radicals, that is, $f(x)$ splits over some extension E/F satisfying $F = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_m = E$ with $F_i = F_{i-1}(\alpha_i)$ satisfying $\alpha_i^{d_i} \in F_{i-1}$ for some d_i for each i . By Lemma 10.1, without loss of generality we may replace E/F by its normal closure N/F to get a finite Galois extension that is radical. Thus E/F is the splitting field of some $\tilde{f}(x) \in F[x]$. Take $n = \prod_{i=1}^m d_i$. Let L/E be the splitting field of $x^n - 1$ over E and $\zeta_n \in L$ a primitive n th root of unity. Set $K = F(\zeta_n)$, and we have $L = E(\zeta_n) = KE$. For each i , define $K_i = KF_i = F_i(\zeta_n)$. Then we have $F \subseteq F(\zeta_n) = K = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_m = F_m(\zeta_n) = L$. Since $F_i = F_{i-1}(\alpha_i)$, we have $K_i = K_{i-1}(\alpha_i)$, where $\alpha_i^{d_i} \in F_{i-1} \subseteq K_{i-1}$, and $\zeta_n \in K_{i-1}$, so $\zeta_{d_i} = \zeta_n^{n/d_i} \in K_{i-1}$. Therefore, by Theorem 9.2, K_i/K_{i-1} is a cyclic Galois extension. But note that L is the splitting field of $\tilde{f}(x)(x^n - 1)$ over F (and thus over each K_i). Hence L/F (also each L/K_i) is Galois. We have

$$\text{Gal}_F(L) \supseteq \text{Gal}_{K_0}(L) \supseteq \text{Gal}_{K_1}(L) \supseteq \cdots \supseteq \text{Gal}_{K_m}(L) = \{1\}$$

by the fundamental theorem. Since K_i/K_{i-1} is Galois, by Theorem 8.4, $\text{Gal}_{K_i}(L)$ is normal in $\text{Gal}_{K_{i-1}}(L)$, and we have $\text{Gal}_{K_{i-1}}(L)/\text{Gal}_{K_i}(L) \cong \text{Gal}_{K_{i-1}}(K_i)$ which is cyclic and thus abelian. Furthermore, $\text{Gal}_F(L)/\text{Gal}_{K_0}(L) = \text{Gal}_F(L)/\text{Gal}_K(L) \cong \text{Gal}_F(K) \subseteq (\mathbb{Z}_n)^*$ (isomorphically) which is abelian. Thus the chain above satisfies the definition to say $\text{Gal}_F(L)$ is solvable. Let $\tilde{E}$ be the splitting field of $\tilde{f}(x)$, which is a subfield of L . Since $\tilde{E}/F$ is a finite Galois extension, by Theorem 8.4, $\text{Gal}(f) = \text{Gal}_F(\tilde{E}) \cong \text{Gal}_F(L)/\text{Gal}_{\tilde{E}}(L)$ which is a quotient of a solvable group and thus solvable by Theorem 6.3. ■

Proposition 10.4

Let $f(x) \in \mathbb{Q}[x]$ be an irreducible polynomial of prime degree p . If $f(x)$ contains exactly two nonreal roots in $\mathbb{C}$, then $\text{Gal}(f) \cong S_p$.

Proof:

Recall that S_n is generated by an n -cycle and 2-cycle. Thus, to show $\text{Gal}(f) \cong S_p$, it suffices to find a p -cycle and a 2-cycle in $\text{Gal}(f)$. Since $\deg(f) = p$, $\text{Gal}(f)$ is a subgroup of S_p . Let α be a root of f . Since $f(x)$ is irreducible of degree p , we have $[\mathbb{Q}(\alpha) : \mathbb{Q}] = \deg(f) = p$, so $p \mid |\text{Gal}(f)|$, so by Cauchy's theorem, $\text{Gal}(f)$ contains an element of order p , which is a p -cycle. On the other hand, the map σ given by complex conjugation will fix the real roots of f and swap the two nonreal roots of f , and so is in $\text{Gal}(f)$ and has order 2, which is a 2-cycle in $\text{Gal}(f)$. ■

Example

Consider $f(x) = x^5 + 2x^3 - 24x - 2 \in \mathbb{Q}[x]$. By Eisenstein's criterion with $p = 2$, this is irreducible. Since $f(-1) = 19$, $f(1) = -23$, $\lim_{x \rightarrow \infty} f(x) \rightarrow \infty$ and $\lim_{x \rightarrow -\infty} f(x) = -\infty$ (by what the graph of a degree 5 polynomial looks like), which implies there are at least 3 real roots of $f(x)$ (again by the shape of the graph of a degree 5 polynomial). Let $\alpha_1, \dots, \alpha_5$ be the roots of $f(x)$. By considering the coefficients of x^4 and x^3 in terms of $f(x)$, we have $\sum_{i=1}^5 \alpha_i = 0$ and $\sum_{i < j} \alpha_i \alpha_j = 2$. From the first sum, we get $0 = (\sum_{i=1}^5 \alpha_i)^2 = 2 \sum_{i < j} \alpha_i \alpha_j$, so $\sum_{i=1}^5 \alpha_i^2 = -4$, so not all roots of $f(x)$ are real. Thus since $f(x)$ has at least 3 real roots, it has exactly 3, with 2 nonreal roots (by the conjugate root theorem).

By Proposition 10.4, we thus have $\text{Gal}(f) \cong S_5$, which is not solvable as seen previously, so by Theorem 10.3, $f(x) = x^5 + 2x^3 - 24x - 2$ over $\mathbb{Q}$ is not solvable by radicals.

Theorem (Abel-Ruffini)

For each $n \geq 5 \in \mathbb{N}$, there exists a polynomial $f(x) \in \mathbb{Q}[x]$ such that $\deg(f) = n$ that is not solvable by radicals. ■

Proposition (A10Q2b, Casus Irreducibilis)

If $[E : \mathbb{Q}] = 3$ with E real, $E/\mathbb{Q}$ is not radical.

Proof:

If we assume that E is radical, our chain is $\mathbb{Q} \subseteq \mathbb{Q}(\alpha) = E$ with $\alpha^d \in \mathbb{Q}$, since E can have no nontrivial intermediate subfields. Now, let $p(x)$ be the minimal polynomial of α over $\mathbb{Q}$. So, $p(x) \mid x^d - \alpha^d$. Let β be some other root of $p(x)$. Then $\beta^d = \alpha^d$ by the previous divisibility. Set $\mu = \beta/\alpha$. Then $\mu^d = 1$, so μ is a d th root of unity. Then we have $\mu \in E$ because $\beta \in E$ (by normality of $E/\mathbb{Q}$ since $\alpha \in E$). Because $p(x)$ has three distinct roots in E by separability, this implies that we get three distinct d th roots of unity in E , which contradicts the fact E is real (since the only real roots of unity are ± 1). Therefore, $E/\mathbb{Q}$ is not radical. ■

This completes the content of the course. Some bonus stuff:

Almost all polynomials $f(x)$ of degree n have $\text{Gal}(f) \cong S_n$. More precisely, if $E_n(N)$ is the number of monic integer polynomials where each coefficient is less than N whose Galois group is a subset of S_n , and $T_n(N)$ is the number of monic integer polynomials where each coefficient is less than N , then Gallagher showed using the large sieve (a method from number theory) that $\frac{E_n(N)}{T_n(N)} \rightarrow 0$ as $N \rightarrow \infty$. There is a conjecture that $E_n(N) \approx (2N + 1)^{n-1}$.

Chapter 11: Additional Topic - Cyclotomic Extensions

For a prime p , we have seen the p th cyclotomic polynomial $\Phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \dots + 1$, which is irreducible in $\mathbb{Q}[x]$. However, for a general $n \geq 2$, $\frac{x^n - 1}{x - 1}$ is not always irreducible, for example, $x^4 - 1 = (x^2 - 1)(x^2 + 1) = (x - 1)(x + 1)(x^2 + 1)$, so $\frac{x^4 - 1}{x - 1} = (x^2 + 1)(x + 1)$ is not irreducible in $\mathbb{Q}[x]$. Thus, if we want to generalize $\Phi_p(x)$, we'll need to go a different direction. For this, note that $\Phi_p(x) = (x - \zeta_p) \dots (x - \zeta_p^{p-1})$ for $\zeta_p = e^{2\pi i/p}$. For each $k = 1, \dots, p - 1$, we have $\gcd(k, p) = 1$, so we can write

$$\Phi_p(x) = \prod_{1 \leq k \leq p, \gcd(k, p)=1} (x - \zeta_p^k)$$

Let $\zeta_n = e^{2\pi i/n}$. For a general $k \in \mathbb{Z}$, the order of ζ_n^k is $\frac{n}{\gcd(n, k)}$. Hence if $\gcd(n, k) = 1$, the order of ζ_n^k is the same as the order of ζ_n . This motivates:

Definition

The n th cyclotomic polynomial, $\Phi_n(x)$ is defined by

$$\Phi_n(x) = \prod_{1 \leq k \leq n, \gcd(n, k)=1} (x - \zeta_n^k)$$

Proposition 11.1

$$x^n - 1 = \prod_{d|n} \Phi_d(x).$$

Theorem 11.2 (Gauss)

$\Phi_n(x)$ is an irreducible integer polynomial.

Theorem 11.3 (Gauss)

$\text{Gal}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n)) \cong (\mathbb{Z}_n)^*$. ■

Definition

For $n \in \mathbb{N}$ and $k \in \mathbb{Z}$ with $\gcd(n, k) = 1$, the field $\mathbb{Q}(\zeta_n) = \mathbb{Q}(\zeta_n^k)$ is called the n th cyclotomic extension over $\mathbb{Q}$.

One reason we care about cyclotomic extensions is:

Theorem 11.4

Let A be a finite abelian group. Then there exists a Galois extension $E/\mathbb{Q}$ with $E \subseteq \mathbb{Q}(\zeta_n)$ and $\text{Gal}_{\mathbb{Q}}(E) \cong A$.

Lemma 11.5

Let p be a prime and $m \in \mathbb{N}$ with $p \nmid m$. Then for $a \in \mathbb{Z}$, p divides $\Phi_m(a)$ if and only if $p \nmid a$ and $a \bmod p$ has order m in the multiplicative group $\mathbb{F}_p^*$.

Because there are infinitely many odd primes, we know there are infinitely many primes of the form $p \equiv 1 \bmod 2$? What about $p \equiv 1 \bmod 4$ or $p \equiv 3 \bmod 4$? A proof similar to Euclid's proof for primes works, if there are finitely many primes $p_1, \dots, p_n$ of the form $3 \bmod 4$ for each $p_i > 3$, then $4p_1 \dots p_n + 3$ is divisible by a prime congruent to $3 \bmod 4$ not in our list. However, this proof does not work for $1 \bmod 4$, there is no guarantee that the prime this proof produces is congruent to $1 \bmod 4$.

Question

For any $m \in \mathbb{N}, k \in \mathbb{Z}$ with $\gcd(k, m) = 1$, are there infinitely primes of the form $p \equiv k \bmod m$? Another way to formulate the question is to ask if for $f(x) = mx + k$, the set of prime divisors of the sequence $\{f(n)\}_{n \in \mathbb{N}}$ is infinite.

Lemma 11.6

If $f(x) \in \mathbb{Z}[x]$ is monic and $\deg(f) \geq 1$, the set of prime divisors of the nonzero integers in the sequence $\{f(n)\}_{n \in \mathbb{N}}$ is infinite.

Using this, one proves:

Theorem 11.7 (Dirichlet)

For $m \in \mathbb{N}, k \in \mathbb{Z}, m \geq 2$ with $\gcd(k, m) = 1$, there are infinitely many primes p of the form $p \equiv k \bmod m$.

Remark

Let $\pi(x)$ be the number of primes less than or equal to x . Then $\pi(x) \sim \frac{x}{\log x}$. Dirichlet proved that for $\gcd(k, m) = 1$, $\pi(x, k, m)$, the number of primes congruent to $k \bmod m$, is $\pi(x, k, m) \sim \frac{1}{\varphi(m)} \pi(x)$.